ResearchQuestionsAnswered

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```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(scales)
library(ggplot2)
library(stringr)
```

```r
brfss1 <- read.csv("completed1_full_brfss.csv",check.names = FALSE)
```

```r
head(brfss1)
```

```
##   row_id MENTHLTH   GENHLTH ADDEPEV3 DECIDE DIFFALON       LSATISFY EMTSUPRT
## 1      1        0      Good       No     No       No Very satisfied   Always
## 2      2        0 Excellent       No     No       No Very satisfied   Always
## 3      3        0 Very good       No     No       No Very satisfied  Usually
## 4      4        0 Excellent       No     No       No Very satisfied   Always
## 5      5        0      Good       No     No      Yes      Satisfied   Rarely
## 6      6        0      Good       No     No       No Very satisfied   Always
##   SDLONELY        HOWSAFE1   _BMI5CAT    _INCOMG1        _SMOKER3 EXERANY2
## 1    Never  Extremely safe     Normal     $15-25k    Never smoked      Yes
## 2    Never  Extremely safe Overweight      >$200k   Former smoker      Yes
## 3    Never  Extremely safe     Normal      >$200k Everyday smoker      Yes
## 4    Never            Safe Overweight     $35-50k    Never smoked      Yes
## 5   Always  Extremely safe     Normal     $15-25k    Never smoked       No
## 6    Never            Safe      Obese $100k-$200k    Never smoked      Yes
##         MARITAL                EDUCA            EMPLOY1 _RACE   _SEX _AGEG5YR
## 1       Widowed High school graduate            Retired White Female    75-79
## 2       Married     College graduate            Retired White   Male      80+
## 3     Partnered         College/tech Employed for Wages White   Male    55-59
```

```
## 4       Married     College graduate            Retired White   Male      80+
## 5 Never married         College/tech     Unable to Work White   Male    45-49
## 6       Married High school graduate Employed for Wages White   Male    50-54
##     _LLCPWT state_name
## 1  261.5255    Alabama
## 2  307.1697    Alabama
## 3 2939.8628    Alabama
## 4  153.5848    Alabama
## 5 1229.6230    Alabama
## 6  451.2199    Alabama
```

```r
names(brfss1) <- sub("^_", "", names(brfss1))
```

```r
round(colMeans(is.na(brfss1))*100,2)
```

```
##     row_id   MENTHLTH    GENHLTH   ADDEPEV3     DECIDE   DIFFALON   LSATISFY
##          0          0          0          0          0          0          0
##   EMTSUPRT   SDLONELY   HOWSAFE1    BMI5CAT    INCOMG1    SMOKER3   EXERANY2
##          0          0          0          0          0          0          0
##    MARITAL      EDUCA    EMPLOY1       RACE        SEX    AGEG5YR     LLCPWT
##          0          0          0          0          0          0          0
## state_name
##          0
```

```r
library(dplyr)
library(stringr)

brfss1 <- brfss1 |>
  mutate(across(where(is.character), str_trim))
```

```r
brfss1$GENHLTH <- factor(
  brfss1$GENHLTH,
  levels = c("Excellent", "Very good", "Good", "Fair", "Poor"),
  ordered = TRUE
)
brfss1$ADDEPEV3  <- factor(brfss1$ADDEPEV3)
brfss1$DECIDE    <- factor(brfss1$DECIDE)
brfss1$DIFFALON  <- factor(brfss1$DIFFALON)
brfss1$LSATISFY <- factor(
  brfss1$LSATISFY,
  levels = c("Very satisfied", "Satisfied", "Dissatisfied", "Very dissatisfied")
)
brfss1$EMTSUPRT <- factor(
  brfss1$EMTSUPRT,
  levels = c("Always", "Usually", "Sometimes", "Rarely", "Never")
)
brfss1$SDLONELY <- factor(
  brfss1$SDLONELY,
  levels = c("Never", "Rarely", "Sometimes", "Usually", "Always"),

)
brfss1$HOWSAFE1 <- factor(
```

```r
  brfss1$HOWSAFE1,
  levels = c("Extremely safe", "Safe", "Unsafe", "Extremely Unsafe")
)
brfss1$BMI5CAT <- factor(
  brfss1$BMI5CAT,
  levels = c("Underweight", "Normal", "Overweight", "Obese")

)
brfss1$INCOMG1 <- factor(
  brfss1$INCOMG1,
  levels = c("<$15k", "$15-25k", "$25-35k", "$35-50k", "$50k-$100k", "$100k-$200k", ">$200k")
)
brfss1$SMOKER3 <- factor(
  brfss1$SMOKER3,
  levels = c("Never smoked", "Former smoker", "Someday smoker", "Everyday smoker")
)
brfss1$EXERANY2  <- factor(brfss1$EXERANY2)
brfss1$MARITAL  <- factor(brfss1$MARITAL)
brfss1$EDUCA  <- factor(brfss1$EDUCA)
brfss1$EMPLOY1  <- factor(brfss1$EMPLOY1)
brfss1$RACE  <- factor(brfss1$RACE)
brfss1$SEX  <- factor(brfss1$SEX)
brfss1$AGEG5YR <- factor(
  brfss1$AGEG5YR,
  levels = c(
    "18-24", "25-29", "30-34", "35-39", "40-44",
    "45-49", "50-54", "55-59", "60-64", "65-69",
    "70-74", "75-79", "80+"
  )
)
brfss1$state_name  <- factor(brfss1$state_name)
```

```r
# We are creating 4 distinct levels based on the days reported
brfss1 <- brfss1 %>%
  mutate(MENTHLTH_CAT = case_when(
    MENTHLTH == 0 ~ "0: Resilient",
    MENTHLTH >= 1 & MENTHLTH <= 13 ~ "1-13: Low/Moderate",
    MENTHLTH >= 14 & MENTHLTH <= 29 ~ "14-29: Frequent",
    MENTHLTH == 30 ~ "30: Chronic"
  ))


brfss1$MENTHLTH_CAT <- factor(brfss1$MENTHLTH_CAT,
                              levels = c("0: Resilient",
                                         "1-13: Low/Moderate",
                                         "14-29: Frequent",
                                         "30: Chronic"),
                              ordered = TRUE)


table(brfss1$MENTHLTH_CAT)
```

```
##
```

```
##       0: Resilient 1-13: Low/Moderate    14-29: Frequent        30: Chronic
##             134033              57994              16740              13035
```

```r
library(survey)
```

```
## Warning: package 'survey' was built under R version 4.5.3

## Loading required package: grid

## Loading required package: Matrix

## Loading required package: survival

##
## Attaching package: 'survey'

## The following object is masked from 'package:graphics':
##
##     dotchart
```

```r
# Creating the survey design using ONLY the weight variable
brfss_design <- svydesign(
  id = ~1,
  weights = ~ LLCPWT,
  data = brfss1
)
```

The hypotheses tested below represent population(US Adults) estimates for the U.S. states, territories, and District of Columbia where social determinants questions were administered.

INCOME

Null Hypothesis (H0): There is no difference in the population mean of poor mental health days across different income groups.

Alternative Hypothesis (H1): The population mean of poor mental health days differs for at least one income group.

For this research question, I conducted a Design-Based One-Way Analysis of Variance (ANOVA) using a Wald Test (via the regTermTest function in R). This test evaluates whether there are statistically significant differences in the mean of a continuous dependent variable (Poor Mental Health Days) across the levels of a categorical independent variable (Income Groups).

```r
model <- svyglm(MENTHLTH ~ INCOMG1, design = brfss_design)

regTermTest(model, "INCOMG1")
```

```
## Wald test for INCOMG1
##  in svyglm(formula = MENTHLTH ~ INCOMG1, design = brfss_design)
## F =  111.827  on  6  and  221795  df: p= < 2.22e-16
```

At a significance level of alpha = 0.05 , the results showed a statistically significant difference in the mean number of poor mental health days across income groups (F (6 , 221795) = 111.83 , p < 0.001) . Since the p-value is less than 0.05, the null hypothesis was rejected, indicating that the mean number of poor mental health days varies significantly across income groups among adults in the selected 33 states/territories in the United States.

After finding a significant overall effect, posthoc pairwise comparisons were performed to identify the specific income groups with significantly different mean poor mental health days.

```r
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
adj_means <- emmeans(model, ~ INCOMG1)

adj_means
```

```
##  INCOMG1     emmean     SE     df lower.CL upper.CL
##  <$15k         7.37 0.2010 221795     6.97     7.76
##  $15-25k       6.05 0.1540 221795     5.75     6.35
##  $25-35k       5.63 0.1280 221795     5.38     5.89
##  $35-50k       5.34 0.1180 221795     5.11     5.57
##  $50k-$100k    4.56 0.0731 221795     4.42     4.70
##  $100k-$200k   3.85 0.0735 221795     3.71     3.99
##  >$200k        3.09 0.1120 221795     2.87     3.31
##
## Confidence level used: 0.95
```

```r
pairs(adj_means, adjust = "bonferroni")
```

```
##  contrast                     estimate    SE     df t.ratio p.value
##  <$15k - ($15-25k)               1.318 0.253 221795   5.206 <0.0001
##  <$15k - ($25-35k)               1.735 0.238 221795   7.274 <0.0001
##  <$15k - ($35-50k)               2.028 0.233 221795   8.695 <0.0001
##  <$15k - ($50k-$100k)            2.809 0.214 221795  13.120 <0.0001
##  <$15k - ($100k-$200k)           3.520 0.214 221795  16.429 <0.0001
##  <$15k - >$200k                  4.280 0.230 221795  18.602 <0.0001
##  ($15-25k) - ($25-35k)           0.417 0.200 221795   2.083  0.7820
##  ($15-25k) - ($35-50k)           0.710 0.194 221795   3.665  0.0052
##  ($15-25k) - ($50k-$100k)        1.491 0.170 221795   8.763 <0.0001
##  ($15-25k) - ($100k-$200k)       2.202 0.170 221795  12.927 <0.0001
##  ($15-25k) - >$200k              2.962 0.190 221795  15.601 <0.0001
##  ($25-35k) - ($35-50k)           0.293 0.174 221795   1.686  1.0000
##  ($25-35k) - ($50k-$100k)        1.075 0.147 221795   7.293 <0.0001
##  ($25-35k) - ($100k-$200k)       1.785 0.148 221795  12.101 <0.0001
##  ($25-35k) - >$200k              2.545 0.170 221795  14.999 <0.0001
##  ($35-50k) - ($50k-$100k)        0.781 0.139 221795   5.632 <0.0001
##  ($35-50k) - ($100k-$200k)       1.492 0.139 221795  10.740 <0.0001
```

```
##  ($35-50k) - >$200k              2.252 0.162 221795  13.878 <0.0001
##  ($50k-$100k) - ($100k-$200k)    0.711 0.104 221795   6.857 <0.0001
##  ($50k-$100k) - >$200k           1.471 0.133 221795  11.031 <0.0001
##  ($100k-$200k) - >$200k          0.760 0.134 221795   5.693 <0.0001
##
## P value adjustment: bonferroni method for 21 tests
```

Estimated mean poor mental health days by income group:-

Estimated Marginal Means (emmeans) were calculated to provide survey weighted predicted means for each group, adjusted for the complex sampling design. This ensures that the means are representative of the actual population proportions in the sampled 33 states/territories

```r
library(dplyr)
library(emmeans)
library(ggplot2)
library(knitr)

# Estimated marginal means
adj_means <- emmeans(model, ~ INCOMG1)

mean_table <- as.data.frame(adj_means) %>%
  select(INCOMG1, emmean, SE, lower.CL, upper.CL) %>%
  mutate(
    emmean = round(emmean, 2),
    SE = round(SE, 2),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2)
  )

kable(
  mean_table,
  col.names = c("Income Group", "Estimated Mean", "SE", "Lower 95% CI", "Upper 95% CI"),
  caption = "Table 1. Estimated mean number of poor mental health days by income group"
)
```

Table 1: Table 1. Estimated mean number of poor mental health days by income group

Income Group	Estimated Mean	SE	Lower 95% CI	Upper 95% CI
<$15k	7.37	0.20	6.97	7.76
$15-25k	6.05	0.15	5.75	6.35
$25-35k	5.63	0.13	5.38	5.89
$35-50k	5.34	0.12	5.11	5.57
$50k-$100k	4.56	0.07	4.42	4.70
$100k-$200k	3.85	0.07	3.71	3.99
>$200k	3.09	0.11	2.87	3.31

```r
# Pairwise comparisons with Bonferroni adjustment
pairwise_table <- as.data.frame(pairs(adj_means, adjust = "bonferroni")) %>%
  select(contrast, estimate, SE, df, p.value) %>%
  mutate(
```

```r
    estimate = round(estimate, 2),
    SE = round(SE, 2),
    p.value = ifelse(p.value < 0.001, "<0.001", round(p.value, 3))
  )

kable(
  pairwise_table,
  col.names = c("Comparison(Income groups)", "Mean Difference", "SE", "df", "Adjusted p-value"),
  caption = "Bonferroni-adjusted pairwise comparisons of income groups"
)
```

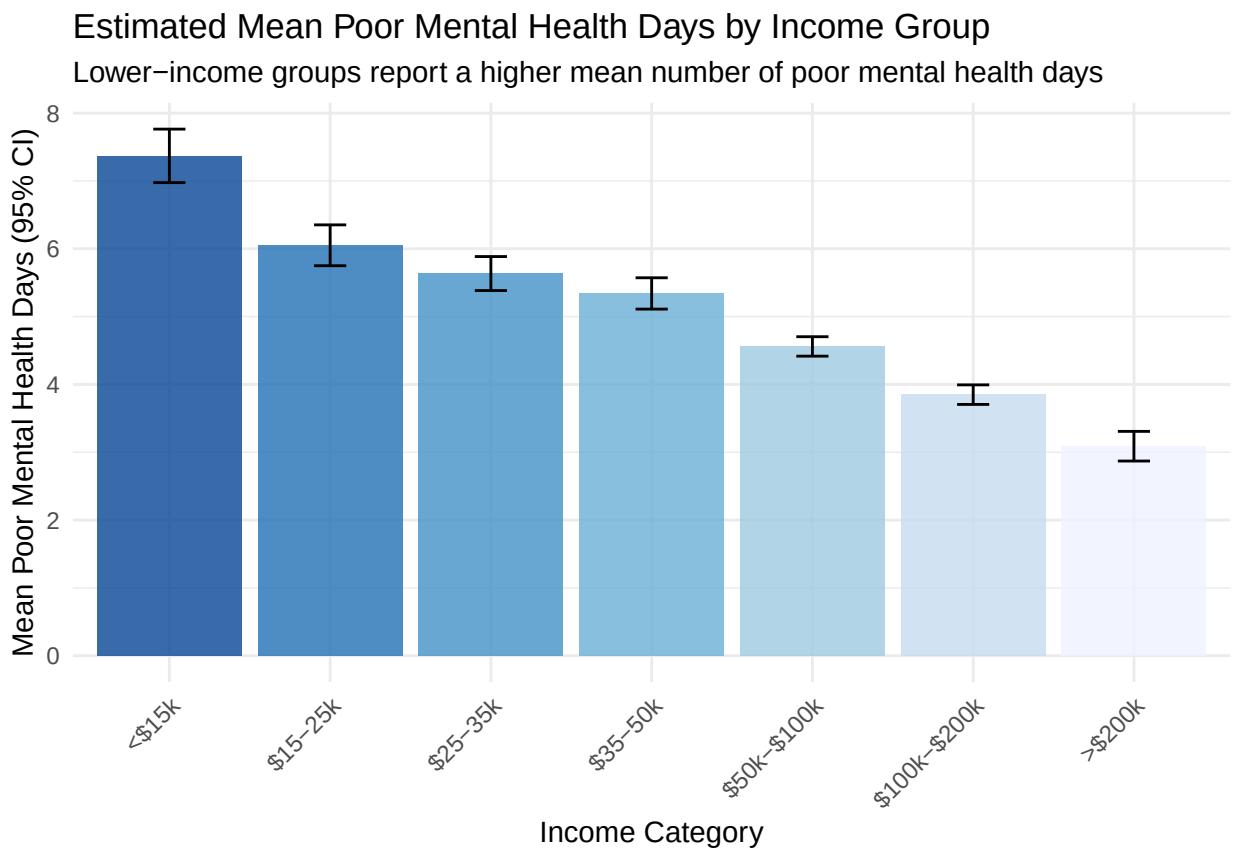
Table 2: Bonferroni-adjusted pairwise comparisons of income groups

Comparison(Income groups)	Mean Difference	SE	df	Adjusted p-value
<$15k - ($15-25k)	1.32	0.25	221795	<0.001
<$15k - ($25-35k)	1.73	0.24	221795	<0.001
<$15k - ($35-50k)	2.03	0.23	221795	<0.001
<$15k - ($50k-$100k)	2.81	0.21	221795	<0.001
<$15k - ($100k-$200k)	3.52	0.21	221795	<0.001
<$15k - >$200k	4.28	0.23	221795	<0.001
($15-25k) - ($25-35k)	0.42	0.20	221795	0.782
($15-25k) - ($35-50k)	0.71	0.19	221795	0.005
($15-25k) - ($50k-$100k)	1.49	0.17	221795	<0.001
($15-25k) - ($100k-$200k)	2.20	0.17	221795	<0.001
($15-25k) - >$200k	2.96	0.19	221795	<0.001
($25-35k) - ($35-50k)	0.29	0.17	221795	1
($25-35k) - ($50k-$100k)	1.07	0.15	221795	<0.001
($25-35k) - ($100k-$200k)	1.79	0.15	221795	<0.001
($25-35k) - >$200k	2.55	0.17	221795	<0.001
($35-50k) - ($50k-$100k)	0.78	0.14	221795	<0.001
($35-50k) - ($100k-$200k)	1.49	0.14	221795	<0.001
($35-50k) - >$200k	2.25	0.16	221795	<0.001
($50k-$100k) - ($100k-$200k)	0.71	0.10	221795	<0.001
($50k-$100k) - >$200k	1.47	0.13	221795	<0.001
($100k-$200k) - >$200k	0.76	0.13	221795	<0.001

Pairwise comparisons indicated that most income groups were significantly different from one another, with lower income groups generally reporting a higher mean number of poor mental health days than higher income groups. Only a few adjacent middle income categories did not differ significantly. Overall, the findings support a clear socioeconomic gradient in poor mental health days.

```r
library(ggplot2)

plot_data <- as.data.frame(adj_means)

p<- ggplot(plot_data, aes(x = INCOMG1, y = emmean, fill = INCOMG1)) +
  geom_bar(stat = "identity", alpha = 0.8) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +

  scale_fill_brewer(palette = "Blues", direction = -1) + # Darker blue = Higher Income
  labs(
```

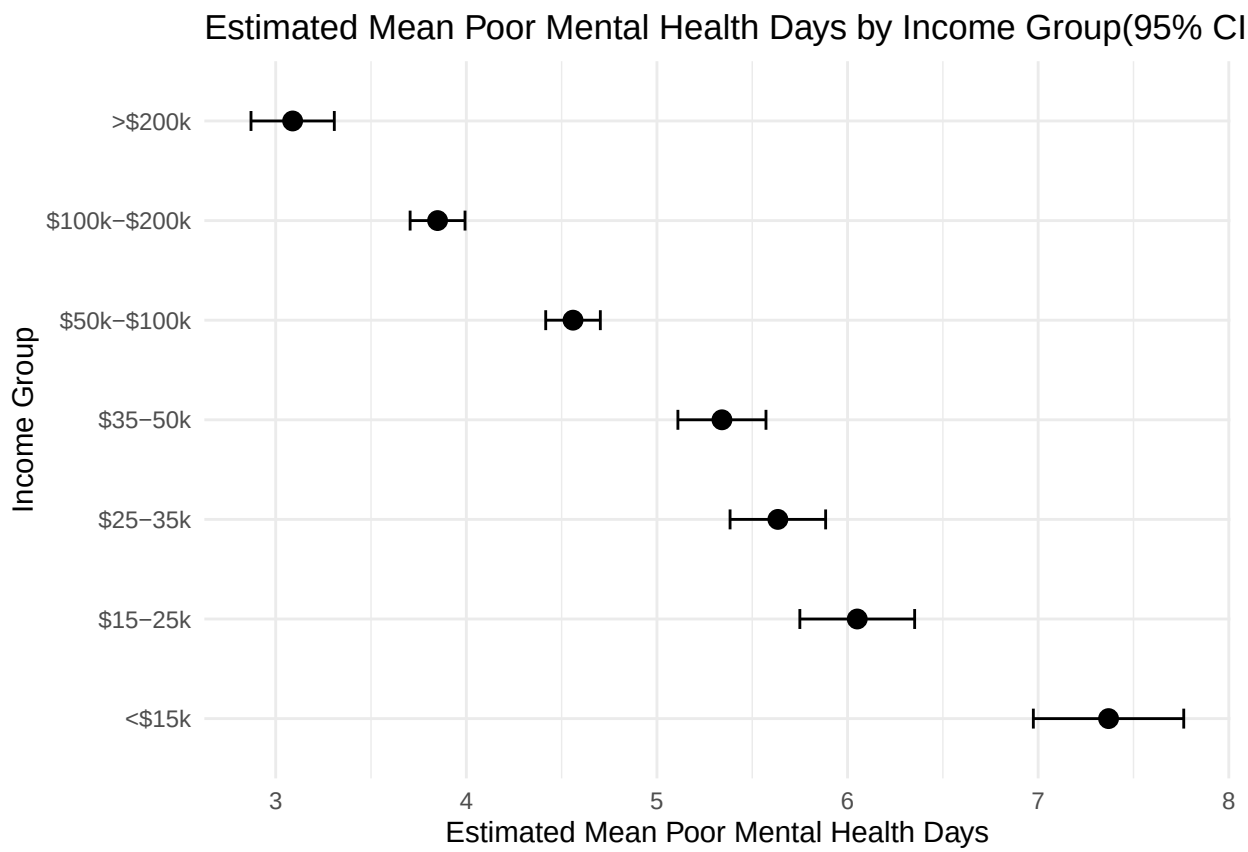
```r
    title = "Estimated Mean Poor Mental Health Days by Income Group",
    subtitle = "Lower-income groups report a higher mean number of poor mental health days",
    x = "Income Category",
    y = "Mean Poor Mental Health Days (95% CI)"
  ) +
  theme_minimal() +
  theme(legend.position = "none", axis.text.x = element_text(angle = 45, hjust = 1))
print(p)
```




```r
ggsave("mean_income_plot.png", width = 8, height = 5, dpi = 300)
```

```r
library(ggplot2)
library(dplyr)

mean_table <- as.data.frame(adj_means) %>%
  select(INCOMG1, emmean, lower.CL, upper.CL)

ggplot(mean_table, aes(x = reorder(INCOMG1, -emmean), y = emmean)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +
  coord_flip() +
  labs(
    title = "Estimated Mean Poor Mental Health Days by Income Group(95% CI)",
    x = "Income Group",
    y = "Estimated Mean Poor Mental Health Days"
```

```
  ) +
  theme_minimal()
```



The estimated mean number of poor mental health days decreased steadily as income increased. Adults in the lowest income group reported the highest mean number of poor mental health days, while those in the highest income group reported the lowest mean. The confidence intervals and pairwise comparisons further support this pattern, showing that most income groups differed significantly from one another. Overall, these findings indicate a clear negative association between income and poor mental health days among adults in the selected 33 states/territories in the United States.

Along with this, we wanted to see the ODD ratio by applying ordinal regression on Mental health category

```
# Ordered Logistic Regression
ordered_model <- svyolr(MENTHLTH_CAT ~ INCOMG1, design = brfss_design)

summary(ordered_model)
```

```
## Call:
## svyolr(MENTHLTH_CAT ~ INCOMG1, design = brfss_design)
##
## Coefficients:
##                        Value Std. Error    t value
## INCOMG1$15-25k     -0.2070560 0.05311580  -3.898199
## INCOMG1$25-35k     -0.2507105 0.05073760  -4.941315
## INCOMG1$35-50k     -0.2994541 0.04897111  -6.114913
## INCOMG1$50k-$100k  -0.3678303 0.04477068  -8.215875
```

```
## INCOMG1$100k-$200k -0.4085067 0.04538035  -9.001840
## INCOMG1>$200k      -0.5666308 0.05317293 -10.656377
##
## Intercepts:
##                                    Value    Std. Error t value
## 0: Resilient|1-13: Low/Moderate     -0.0695   0.0414    -1.6763
## 1-13: Low/Moderate|14-29: Frequent   1.3786   0.0427    32.2543
## 14-29: Frequent|30: Chronic          2.3450   0.0451    52.0003
```

Odd ratios = exp(coefficient)

```r
library(dplyr)
library(stringr)

coef_table <- summary(ordered_model)$coefficients

or_table_income <- data.frame(
  Variable = rownames(coef_table),
  Coefficient = coef_table[, "Value"],
  SE = coef_table[, "Std. Error"],
  t_value = coef_table[, "t value"]
) %>%
  filter(str_detect(Variable, "^INCOMG1")) %>%
  mutate(
    `Income Group` = c("$15-25k", "$25-35k", "$35-50k", "$50k-$100k", "$100k-$200k", ">$200k"),
    OR = round(exp(Coefficient), 2),
    `Change in Odds` = ifelse(
      exp(Coefficient) < 1,
      paste0(round((1 - exp(Coefficient)) * 100, 1), "% lower odds"),
      paste0(round((exp(Coefficient) - 1) * 100, 1), "% higher odds")
    ),
    SE = round(SE, 3),
    t_value = round(t_value, 2)
  ) %>%
  select(`Income Group`, OR, SE, t_value, `Change in Odds`)

or_table_income
```

```
##                     Income Group   OR    SE t_value   Change in Odds
## INCOMG1$15-25k          $15-25k 0.81 0.053   -3.90 18.7% lower odds
## INCOMG1$25-35k          $25-35k 0.78 0.051   -4.94 22.2% lower odds
## INCOMG1$35-50k          $35-50k 0.74 0.049   -6.11 25.9% lower odds
## INCOMG1$50k-$100k    $50k-$100k 0.69 0.045   -8.22 30.8% lower odds
## INCOMG1$100k-$200k  $100k-$200k 0.66 0.045   -9.00 33.5% lower odds
## INCOMG1>$200k            >$200k 0.57 0.053  -10.66 43.3% lower odds
```

Compared with adults earning less than 15k, all higher income groups had lower odds of being in a worse mental health category. The odds decreased progressively as income increased. For example, adults earning 15k-25k had 19% lower odds of being in a worse mental health category, whereas those earning more than $200k had approximately 43% lower odds. Overall, these results indicate that higher income is associated with lower odds of more severe poor mental health.

The relationship between income and mental health was examined using both a survey-weighted analysis of variance and an ordered logistic regression model. Both approaches consistently indicated that income is significantly associated with mental health outcomes.

OR =1 :- meaning no difference OR >1 :- more likely(higher odds) OR <1 :- Less likely (lower odds)

AGE

The original 13 five-year age categories from the BRFSS dataset were collapsed into four broader life-stage groups (18-34, 35-49, 50-64, and 65+). Doing this improves the statistical power of the analysis and allows for a clearer interpretation of how mental health burden shifts across the lifespan.

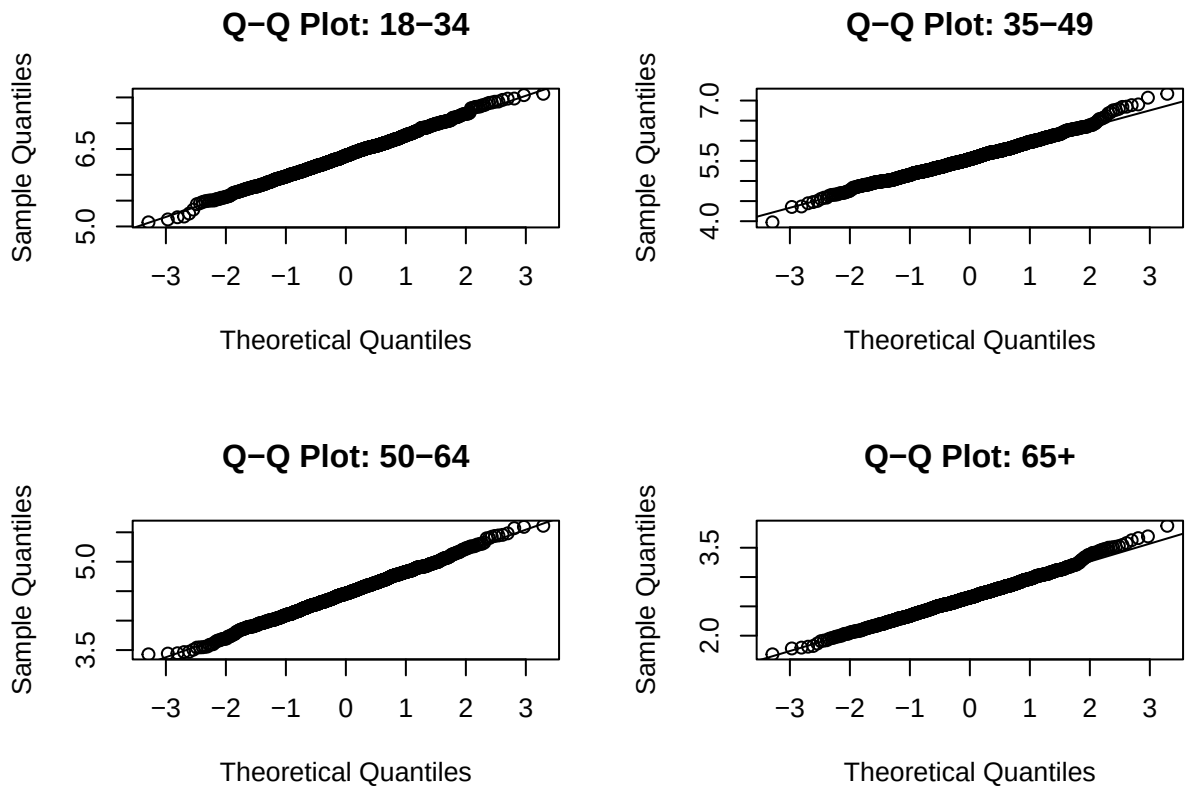
Null Hypothesis (H0):- The population mean of poor mental health days is equal across all four age groups. Alternative Hypothesis (H1) :- At least one age group has a population mean of poor mental health days that differs from the others.

```
brfss1$AGE4 <- dplyr::case_when(
  brfss1$AGEG5YR %in% c("18-24", "25-29", "30-34") ~ "18-34",
  brfss1$AGEG5YR %in% c("35-39", "40-44", "45-49") ~ "35-49",
  brfss1$AGEG5YR %in% c("50-54", "55-59", "60-64") ~ "50-64",
  brfss1$AGEG5YR %in% c("65-69", "70-74", "75-79", "80+") ~ "65+"
)
```

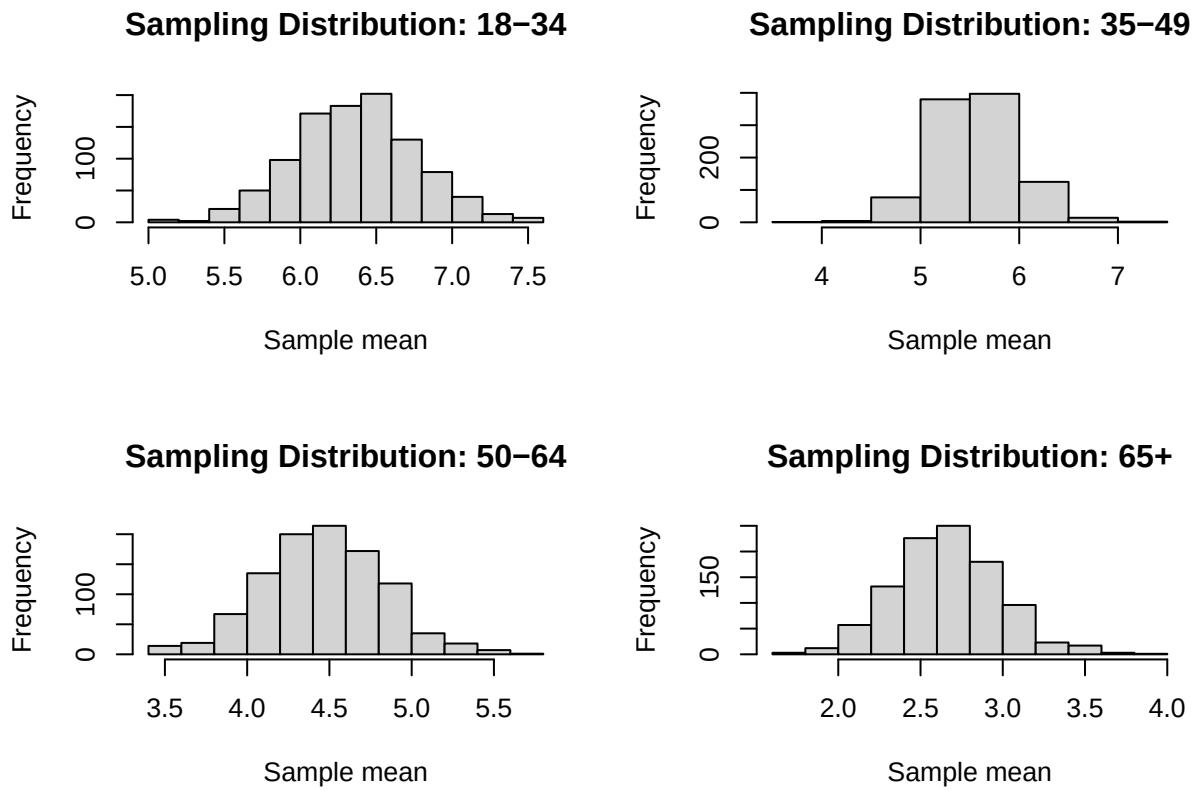
```
brfss1$AGE4 <- factor(brfss1$AGE4, levels = c("18-34", "35-49", "50-64", "65+"))
```

```
brfss_design <- update(brfss_design, AGE4 = brfss1$AGE4)
```

```
set.seed(1)

groups <- levels(brfss1$AGE4)

par(mfrow = c(2, 2))

for (g in groups) {
  x <- subset(brfss1, AGE4 == g)$MENTHLTH
  x <- x[!is.na(x)]

  sample_means <- replicate(1000, mean(sample(x, 500, replace = TRUE)))

  qqnorm(sample_means, main = paste("Q-Q Plot:", g))
  qqline(sample_means)
}
```




```r
set.seed(1)

groups <- levels(brfss1$AGE4)

par(mfrow = c(2, 2))

for (g in groups) {
  x <- subset(brfss1, AGE4 == g)$MENTHLTH
  x <- x[!is.na(x)]

  sample_means <- replicate(1000, mean(sample(x, 500, replace = TRUE)))

  hist(sample_means,
       main = paste("Sampling Distribution:", g),
       xlab = "Sample mean")
}
```



To assess the applicability of the Central Limit Theorem, a resampling procedure was performed within each age group. Repeated samples of size 500 were drawn, and Q-Q plots of the resulting sample means showed that the sampling distributions were approximately normal across all age groups.

To evaluate the relationship between age and mental health, a design-based One-Way ANOVA was performed using a Wald test.

```
age4_model <- svyglm(MENTHLTH ~ AGE4, design = brfss_design)



regTermTest(age4_model, ~ AGE4)
```

```
## Wald test for AGE4
##  in svyglm(formula = MENTHLTH ~ AGE4, design = brfss_design)
## F =  502.717  on  3  and  221798  df: p= < 2.22e-16
```

The survey weighted Wald test showed a significant overall effect of age group on mean poor mental health days (F=502.72,p<0.001). This indicates that at least one age group has a different mean number of poor mental health days.

Because the overall age group effect was statistically significant, post hoc pairwise comparisons were conducted to determine which age groups differed significantly in their mean number of poor mental health days.

```r
library(emmeans)

age4_emm <- emmeans(age4_model, ~ AGE4)
age4_emm
```

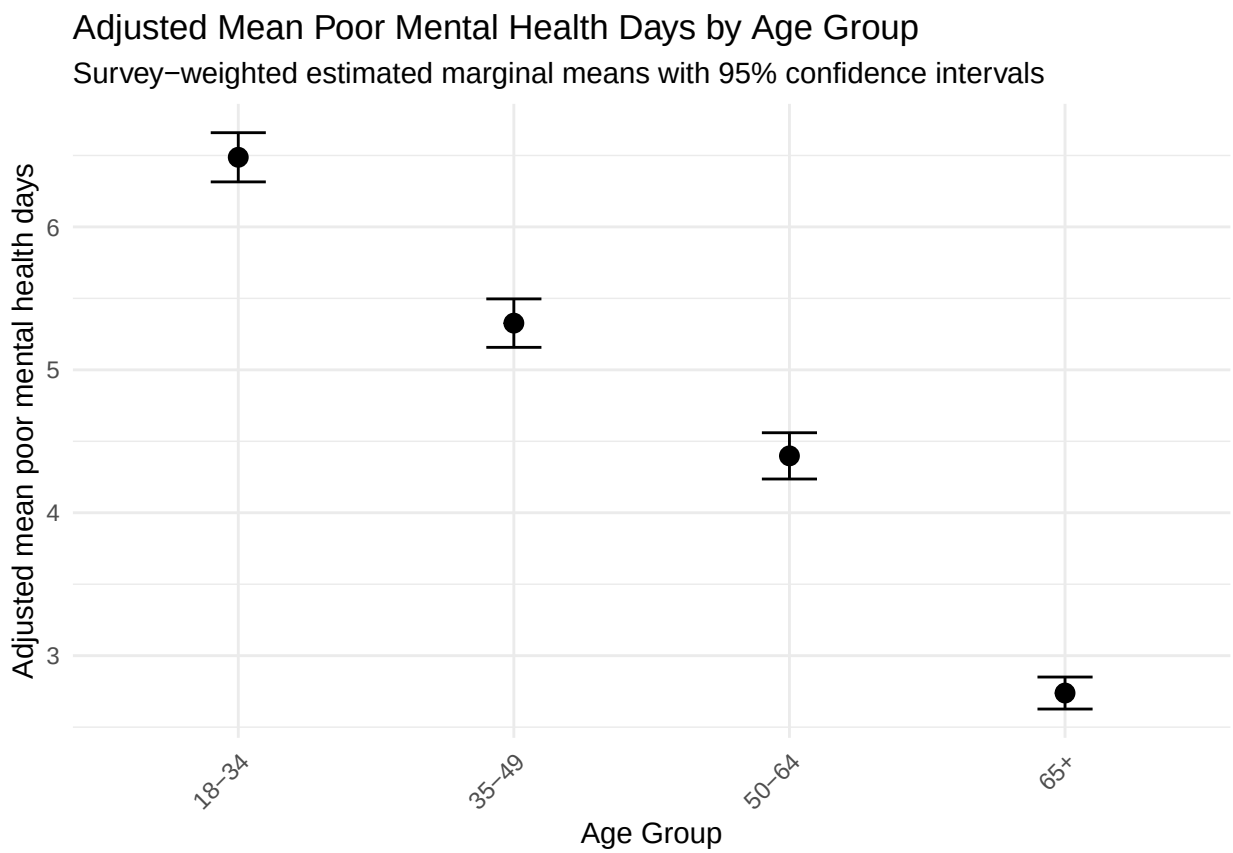
```
##  AGE4  emmean     SE     df lower.CL upper.CL
##  18-34   6.49 0.0878 221798     6.32     6.66
##  35-49   5.33 0.0865 221798     5.16     5.50
##  50-64   4.40 0.0826 221798     4.24     4.56
##  65+     2.74 0.0571 221798     2.63     2.85
##
## Confidence level used: 0.95
```

```r
pairs(age4_emm, adjust = "bonferroni")
```

```
##  contrast          estimate    SE     df t.ratio p.value
##  (18-34) - (35-49)    1.160 0.123 221798   9.415 <0.0001
##  (18-34) - (50-64)    2.089 0.121 221798  17.331 <0.0001
##  (18-34) - (65+)      3.749 0.105 221798  35.786 <0.0001
##  (35-49) - (50-64)    0.929 0.120 221798   7.769 <0.0001
##  (35-49) - (65+)      2.588 0.104 221798  24.983 <0.0001
##  (50-64) - (65+)      1.659 0.100 221798  16.532 <0.0001
##
## P value adjustment: bonferroni method for 6 tests
```

Post hoc pairwise comparisons using Bonferroni adjustment indicated that every age category differed significantly from every other age category ($p<0.001$). Compared with adults aged 35-49, those aged 18-34 reported, on average, 1.16 more poor mental health days. Compared with those aged 50-64 and 65+, the 18-34 group reported 2.09 and 3.75 more poor mental health days, respectively. Similarly, the 35-49 and 50-64 groups each had significantly higher mean poor mental health days than the 65+ group. Overall, the findings suggest that mean poor mental health days decline progressively with age.

```r
library(ggplot2)

age_means_df <- as.data.frame(age4_emm)

q<-ggplot(age_means_df, aes(x = AGE4, y = emmean)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +
  labs(
    x = "Age Group",
    y = "Adjusted mean poor mental health days",
    title = "Adjusted Mean Poor Mental Health Days by Age Group",
    subtitle = "Survey-weighted estimated marginal means with 95% confidence intervals"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(q)
```




```
ggsave("mean_age_plot.png", width = 8, height = 5, dpi = 300)
```

The estimated marginal means show a clear decreasing trend, where younger adults (1834 years) report the highest number of poor mental health days, while older adults (65+ years) report substantially fewer poor mental health days.

```
library(emmeans)
library(dplyr)
library(knitr)

# Pairwise comparisons
age_pairs <- pairs(age4_emm, adjust = "bonferroni")

age_pairs_df <- as.data.frame(age_pairs)


age_pairs_table <- age_pairs_df %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, t.ratio, p.value)
```

```r
kable(
  age_pairs_table,
  caption = "Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Age G
  col.names = c("Age Group Comparison", "Mean Difference", "SE", "df", "t Ratio", "p-value"),
  align = "lccccc"
)
```

Table 3: Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Age Group

Age Group Comparison	Mean Difference	SE	df	t Ratio	p-value
(18-34) - (35-49)	1.160	0.123	221798	9.415	<0.001
(18-34) - (50-64)	2.089	0.121	221798	17.331	<0.001
(18-34) - (65+)	3.749	0.105	221798	35.786	<0.001
(35-49) - (50-64)	0.929	0.120	221798	7.769	<0.001
(35-49) - (65+)	2.588	0.104	221798	24.983	<0.001
(50-64) - (65+)	1.659	0.100	221798	16.532	<0.001

```r
age_pairs
```

```
##  contrast          estimate    SE     df t.ratio p.value
##  (18-34) - (35-49)    1.160 0.123 221798   9.415 <0.0001
##  (18-34) - (50-64)    2.089 0.121 221798  17.331 <0.0001
##  (18-34) - (65+)      3.749 0.105 221798  35.786 <0.0001
##  (35-49) - (50-64)    0.929 0.120 221798   7.769 <0.0001
##  (35-49) - (65+)      2.588 0.104 221798  24.983 <0.0001
##  (50-64) - (65+)      1.659 0.100 221798  16.532 <0.0001
##
## P value adjustment: bonferroni method for 6 tests
```

```r
age_pairs_ci <- confint(age_pairs)

age_pairs_ci_df <- as.data.frame(age_pairs_ci)
```

```r
sig_age_pairs_table_ci <- age_pairs_df %>%
  mutate(
    lower.CL = age_pairs_ci_df$lower.CL,
    upper.CL = age_pairs_ci_df$upper.CL
  ) %>%
  filter(p.value < 0.05) %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    lower.CL = round(lower.CL, 3),
    upper.CL = round(upper.CL, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, lower.CL, upper.CL, t.ratio, p.value)
```

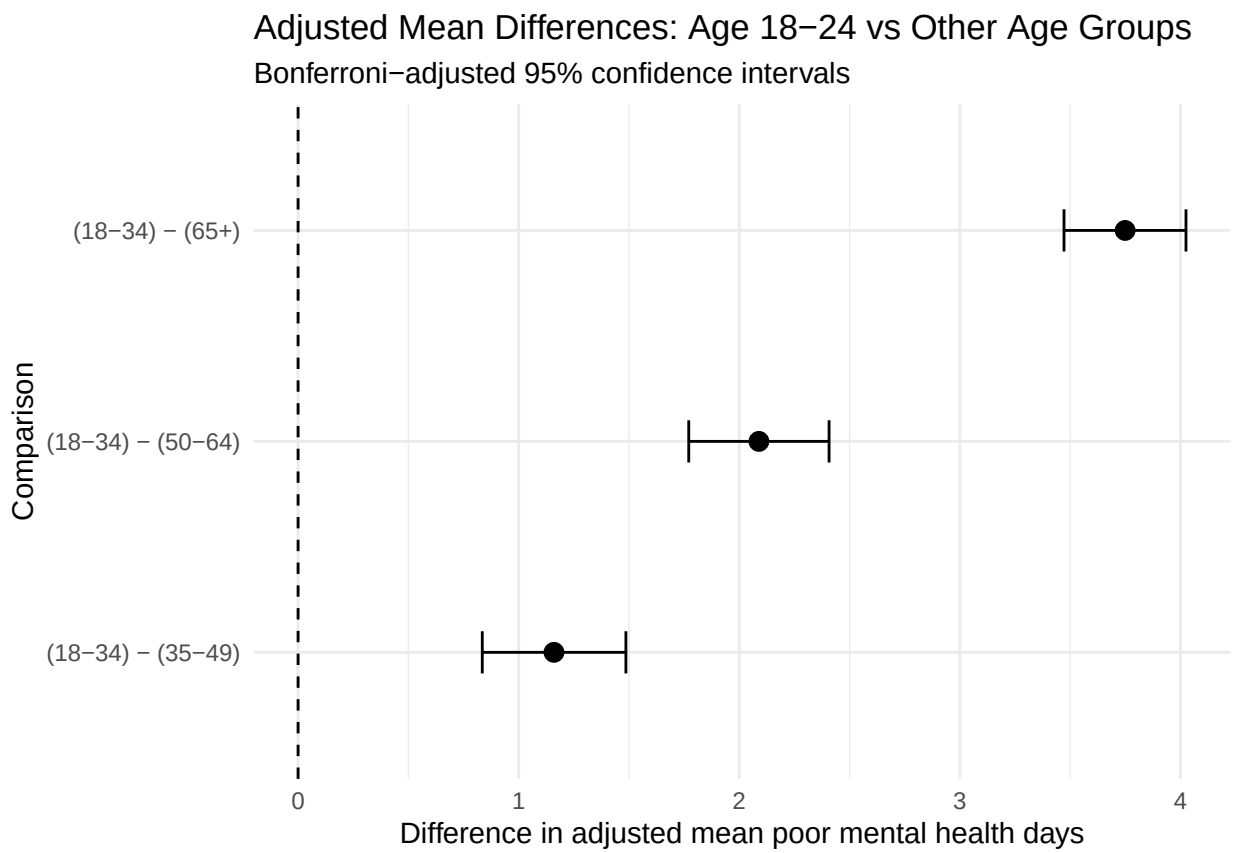
```r
kable(
  sig_age_pairs_table_ci,
  caption = "Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Da
  col.names = c("Comparison", "Mean Difference", "SE", "df", "CI Lower", "CI Upper", "t Ratio", "p-value
  align = "lcccccc"
)
```

Table 4: Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Age Group

Comparison	Mean Difference	SE	df	CI Lower	CI Upper	t Ratio	p-value
(18-34) - (35-49)	1.160	0.123	221798	0.835	1.486	9.415	<0.001
(18-34) - (50-64)	2.089	0.121	221798	1.771	2.407	17.331	<0.001
(18-34) - (65+)	3.749	0.105	221798	3.472	4.025	35.786	<0.001
(35-49) - (50-64)	0.929	0.120	221798	0.613	1.244	7.769	<0.001
(35-49) - (65+)	2.588	0.104	221798	2.315	2.861	24.983	<0.001
(50-64) - (65+)	1.659	0.100	221798	1.395	1.924	16.532	<0.001

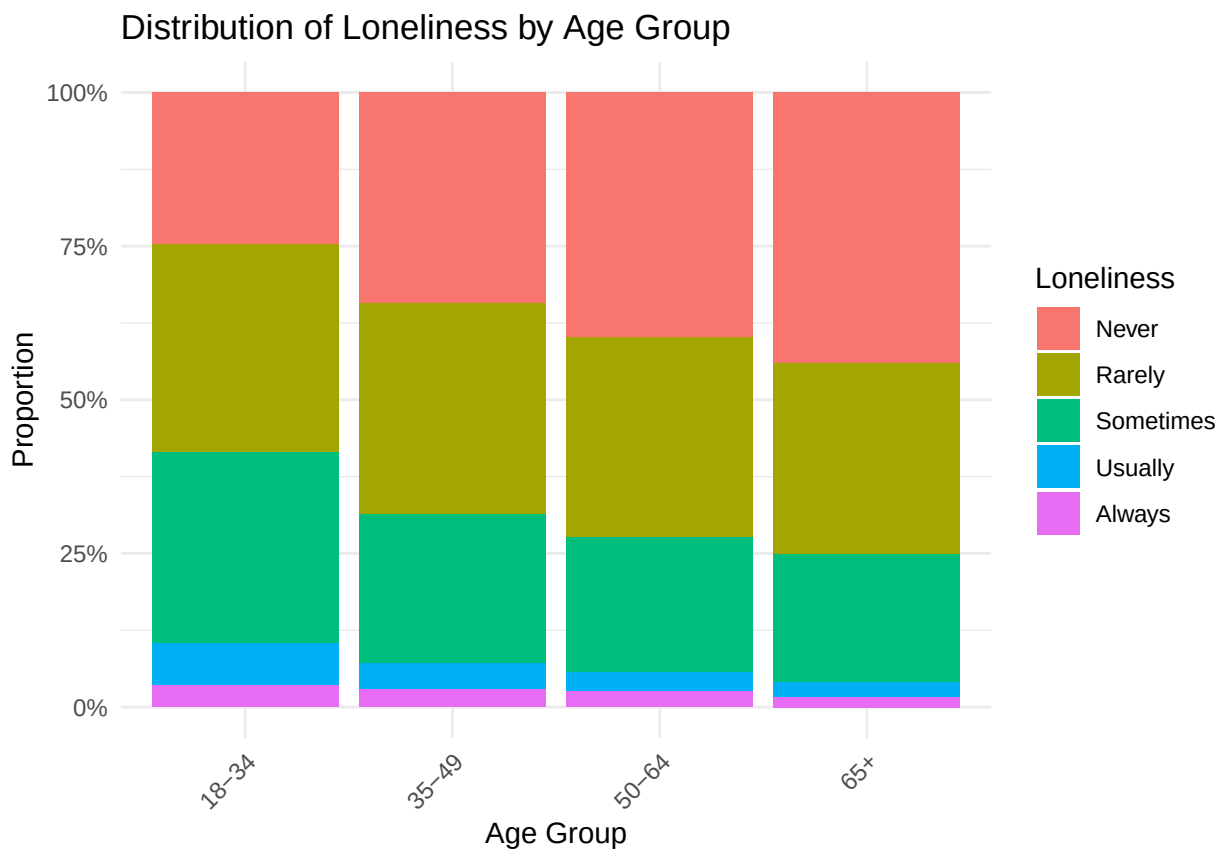
```r
library(ggplot2)
library(dplyr)

plot_df <- sig_age_pairs_table_ci %>%
  filter(grepl("^\\(18-34\\)", contrast) | grepl("^18-34", contrast))

ggplot(plot_df, aes(x = estimate, y = reorder(contrast, estimate))) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = `lower.CL`, xmax = `upper.CL`), height = 0.2) +
  geom_vline(xintercept = 0, linetype = 2) +
  labs(
    x = "Difference in adjusted mean poor mental health days",
    y = "Comparison",
    title = "Adjusted Mean Differences: Age 18-24 vs Other Age Groups",
    subtitle = "Bonferroni-adjusted 95% confidence intervals"
  ) +
  theme_minimal()
```

```
## Warning: ‘geom_errorbarh()‘ was deprecated in ggplot2 4.0.0.
## i Please use the ‘orientation‘ argument of ‘geom_errorbar()‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



The pairwise comparison plot shows that individuals aged 18-34 report significantly more poor mental health days compared to all other age groups. All estimated differences are positive and their confidence intervals do not include zero, indicating statistically significant differences. Furthermore, the magnitude of the difference increases with age, with the largest gap observed between the youngest and oldest age groups. This suggests a strong age-related gradient in mental health outcomes.

```
library(ggplot2)
library(dplyr)

ggplot(
  brfss1 %>% filter(!is.na(AGE4), !is.na(SDLONELY)),
  aes(x = AGE4, fill = SDLONELY)
) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    x = "Age Group",
    y = "Proportion",
    fill = "Loneliness",
    title = "Distribution of Loneliness by Age Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



To further investigate the relative risk across age groups, a survey-weighted ordinal logistic regression was performed. The model used 'Mental Health Category' as an ordinal dependent variable (Resilient, Low/Moderate, Frequent, Chronic).

```
#Ordered Logistic Regression
ordered_model_age <- svyolr(MENTHLTH_CAT ~ AGE4, design = brfss_design)

summary(ordered_model_age)
```

```
## Call:
## svyolr(MENTHLTH_CAT ~ AGE4, design = brfss_design)
##
## Coefficients:
##                Value Std. Error   t value
## AGE435-49 -0.3412034 0.02587019 -13.18906
## AGE450-64 -0.7016298 0.02684976 -26.13170
## AGE465+   -1.3417735 0.02674439 -50.17028
##
## Intercepts:
##                                    Value    Std. Error t value
## 0: Resilient|1-13: Low/Moderate     -0.2654   0.0176   -15.0931
## 1-13: Low/Moderate|14-29: Frequent   1.2444   0.0182    68.2540
## 14-29: Frequent|30: Chronic          2.2193   0.0226    98.1658
```

```r
coef_table_age <- coef(summary(ordered_model_age))
```

```r
or_table_age <- data.frame(
  Variable = rownames(coef_table_age),
  Coefficient = coef_table_age[, "Value"],
  SE = coef_table_age[, "Std. Error"],
  t_value = coef_table_age[, "t value"]
) %>%
  filter(str_detect(Variable, "^AGE4")) %>%
  mutate(
    `Age Group` = str_remove(Variable, "^AGE4"),
    OR = round(exp(Coefficient), 2),
    p_raw = 2 * pnorm(abs(t_value), lower.tail = FALSE),
    p_adj = pmin(p_raw * 3, 1),
    `P Value Adj` = format.pval(p_adj, eps = .001, digits = 3),
    `Change in Odds` = ifelse(
      exp(Coefficient) < 1,
      paste0(round((1 - exp(Coefficient)) * 100, 1), "% lower odds"),
      paste0(round((exp(Coefficient) - 1) * 100, 1), "% higher odds")
    ),
    SE = round(SE, 3),
    t_value = round(t_value, 2)
  ) %>%
  select(`Age Group`, OR, SE, t_value, `P Value Adj`, `Change in Odds`)

or_table_age
```

```
##           Age Group   OR    SE t_value P Value Adj   Change in Odds
## AGE435-49     35-49 0.71 0.026  -13.19      <0.001 28.9% lower odds
## AGE450-64     50-64 0.50 0.027  -26.13      <0.001 50.4% lower odds
## AGE465+         65+ 0.26 0.027  -50.17      <0.001 73.9% lower odds
```

```r
or_table_age
```

```
##           Age Group   OR    SE t_value P Value Adj   Change in Odds
## AGE435-49     35-49 0.71 0.026  -13.19      <0.001 28.9% lower odds
## AGE450-64     50-64 0.50 0.027  -26.13      <0.001 50.4% lower odds
## AGE465+         65+ 0.26 0.027  -50.17      <0.001 73.9% lower odds
```

```r
library(dplyr)
library(knitr)

or_table_age %>%
  mutate(
    OR = round(OR, 2),
    SE = round(SE, 3),
    t_value = round(t_value, 2)
  ) %>%
  kable(
    caption = "Odds Ratios for Mental Health Severity by Age Group",
    col.names = c("Age Group", "OR", "SE", "t value", "Adjusted p-value", "Change in Odds"),
    align = "lccccc"
  )
```

Table 5: Odds Ratios for Mental Health Severity by Age Group

	Age Group	OR	SE	t value	Adjusted p-value	Change in Odds
AGE435-49	35-49	0.71	0.026	-13.19	<0.001	28.9% lower odds
AGE450-64	50-64	0.50	0.027	-26.13	<0.001	50.4% lower odds
AGE465+	65+	0.26	0.027	-50.17	<0.001	73.9% lower odds

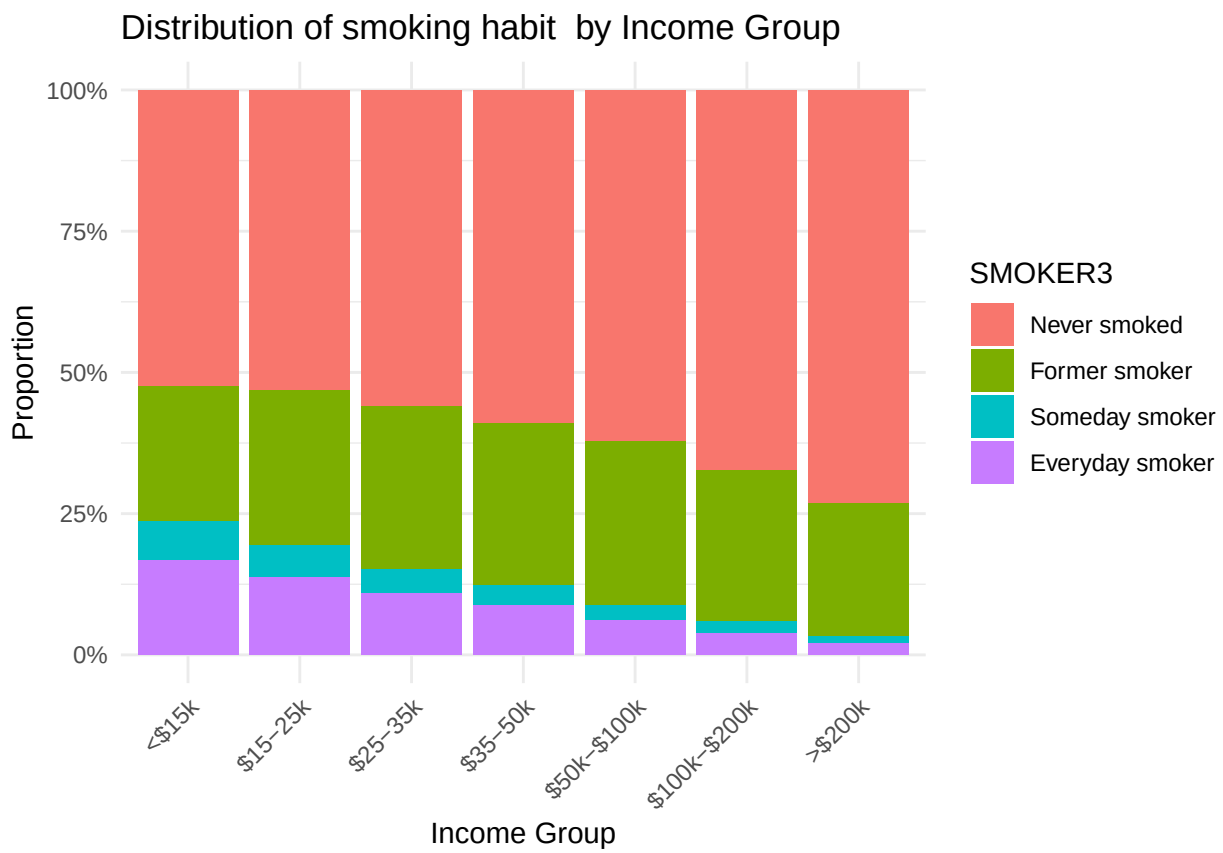
Using adults aged 18-34 as the reference group, the odds of being in a worse mental health category decreased with age. Adults aged 35-49 had 28.9% lower odds, those aged 50-64 had 50.4% lower odds, and those aged 65+ had 73.9% lower odds of worse mental health compared with adults aged 18-34.

SMOKER3

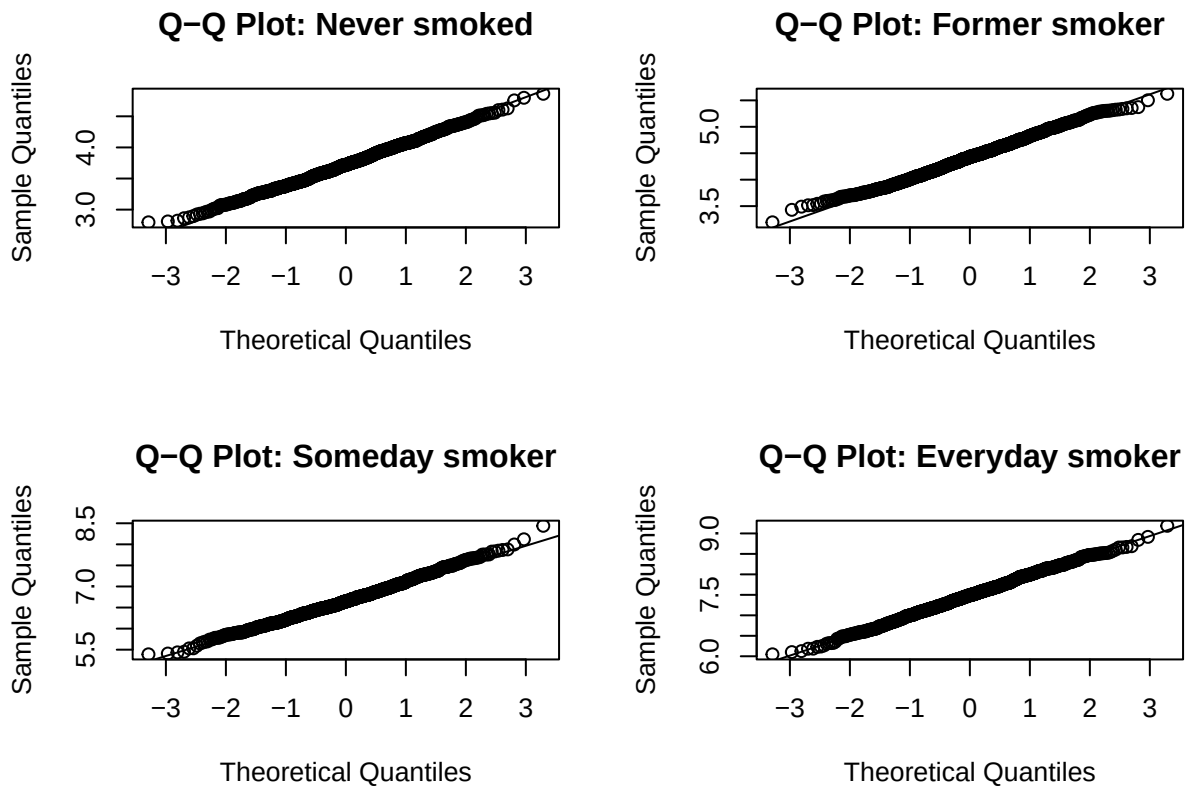
Null Hypothesis (H0): There is no difference in the mean number of poor mental health days across the four smoking categories in the population.

Alternative Hypothesis (H1): At least one smoking category has a significantly different mean number of poor mental health days compared to the others.

```
library(ggplot2)
library(dplyr)

ggplot(
  brfss1 %>% filter(!is.na(INCOMG1), !is.na(SMOKER3)),
  aes(x = INCOMG1, fill = SMOKER3)
) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    x = "Income Group",
    y = "Proportion",
    fill = "SMOKER3",
    title = "Distribution of smoking habit  by Income Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



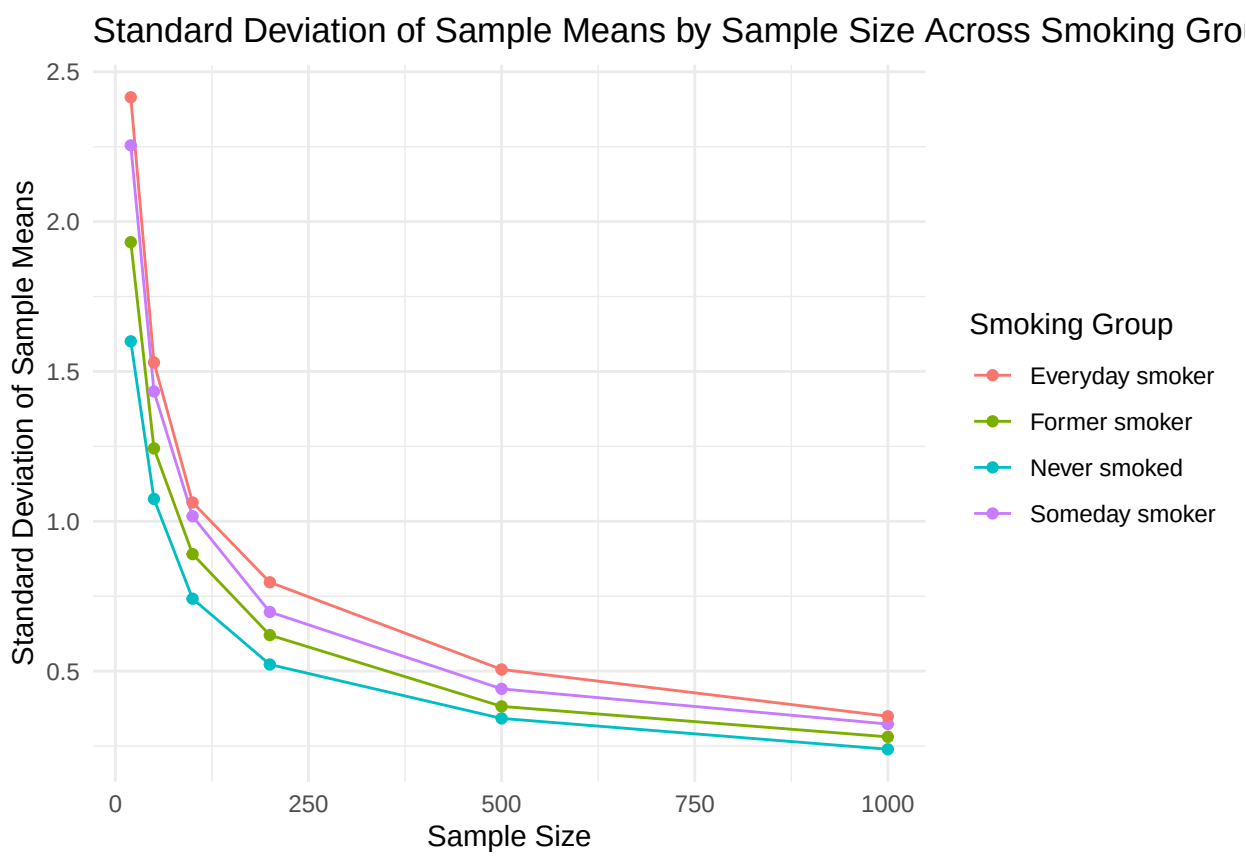

```r
set.seed(1)

groups <- levels(brfss1$SMOKER3)

par(mfrow = c(2, 2))

for (g in groups) {
  x <- subset(brfss1, SMOKER3 == g)$MENTHLTH
  x <- x[!is.na(x)]

  sample_means <- replicate(1000, mean(sample(x, 500, replace = TRUE)))

  qqnorm(sample_means, main = paste("Q-Q Plot:", g))
  qqline(sample_means)
}
```



To validate the applicability of the Central Limit Theorem (CLT), a resampling approach was conducted for each smoking category. Repeated samples were drawn, and the distribution of sample means was examined using both histograms and Q-Q plots. Across all groups, the sampling distributions were approximately symmetric and bell-shaped, and the Q-Q plots closely followed a straight line, indicating approximate normality. These results confirm that, despite the skewness in the original data and residuals, the sampling distribution of the mean is approximately normal. Therefore, the assumptions required for ANOVA-based inference are reasonably satisfied.

```r
set.seed(1)

sample_sizes <- c(20, 50, 100, 200, 500, 1000)
groups <- levels(brfss1$SMOKER3)

results <- data.frame()

for (g in groups) {
  x <- brfss1$MENTHLTH[brfss1$SMOKER3 == g]
  x <- x[!is.na(x)]

  se_vals <- sapply(sample_sizes, function(n) {
    sample_means <- replicate(1000, mean(sample(x, n, replace = TRUE)))
    sd(sample_means)
  })

  temp <- data.frame(
    sample_size = sample_sizes,
```

```r
    sd_of_sample_means = se_vals,
    group = g
  )

  results <- rbind(results, temp)
}

ggplot(results, aes(x = sample_size, y = sd_of_sample_means, color = group)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Standard Deviation of Sample Means by Sample Size Across Smoking Groups",
    x = "Sample Size",
    y = "Standard Deviation of Sample Means",
    color = "Smoking Group"
  ) +
  theme_minimal()
```



To further illustrate the Central Limit Theorem, the standard deviation of the resampled sample means was plotted against increasing sample size. The plot showed that the variability of the sample mean decreased as sample size increased, consistent with the relationship This supports the idea that larger samples produce more stable mean estimates and strengthens the justification for ANOVA-based inference.

```r
model_smoking <- svyglm(MENTHLTH ~ SMOKER3, design = brfss_design)

regTermTest(model_smoking, "SMOKER3")
```

```
## Wald test for SMOKER3
##  in svyglm(formula = MENTHLTH ~ SMOKER3, design = brfss_design)
## F =  178.0263  on  3  and  221798  df: p= < 2.22e-16
```

A survey-weighted Wald test revealed a highly significant relationship between smoking and poor mental health days(F=178.0263, P<0.001). We can reject Ho, this confirms that the number of poor mental health days is not equal across smoking habits, allowing for further post hoc analysis of specific group differences.

```r
library(emmeans)

smoke_emm <- emmeans(model_smoking, ~ SMOKER3)
smoke_emm
```

```
##  SMOKER3         emmean     SE     df lower.CL upper.CL
##  Never smoked      4.26 0.0475 221798     4.17     4.36
##  Former smoker     5.20 0.0870 221798     5.02     5.37
##  Someday smoker    6.77 0.2360 221798     6.31     7.23
##  Everyday smoker   8.11 0.1850 221798     7.75     8.47
##
## Confidence level used: 0.95
```

```r
pairs(smoke_emm, adjust = "bonferroni")
```

```
##  contrast                          estimate     SE     df t.ratio p.value
##  Never smoked - Former smoker        -0.932 0.0991 221798  -9.403 <0.0001
##  Never smoked - Someday smoker       -2.506 0.2410 221798 -10.390 <0.0001
##  Never smoked - Everyday smoker      -3.849 0.1910 221798 -20.167 <0.0001
##  Former smoker - Someday smoker      -1.574 0.2520 221798  -6.246 <0.0001
##  Former smoker - Everyday smoker     -2.917 0.2040 221798 -14.278 <0.0001
##  Someday smoker - Everyday smoker    -1.343 0.3000 221798  -4.475 <0.0001
##
## P value adjustment: bonferroni method for 6 tests
```

Post-hoc pairwise comparisons using Bonferroni adjustment revealed significant differences between all smoking categories . A clear relationship was observed: everyday smokers reported the highest mean number of poor mental health days (M = 8.11, 95% CI [7.75, 8.47]), which was significantly higher than someday smokers (M = 6.77), former smokers (M = 5.20), and never-smokers (M = 4.26).

t.ratio = estimate / SE Estimate: The actual difference in days (e.g., Never smoked minus Everyday smoker = -3.84). SE (Standard Error): The "noise" or uncertainty in the survey data (0.19). Result: A t.ratio far away from zero (either positive or negative) means the "signal" is much stronger than the "noise."

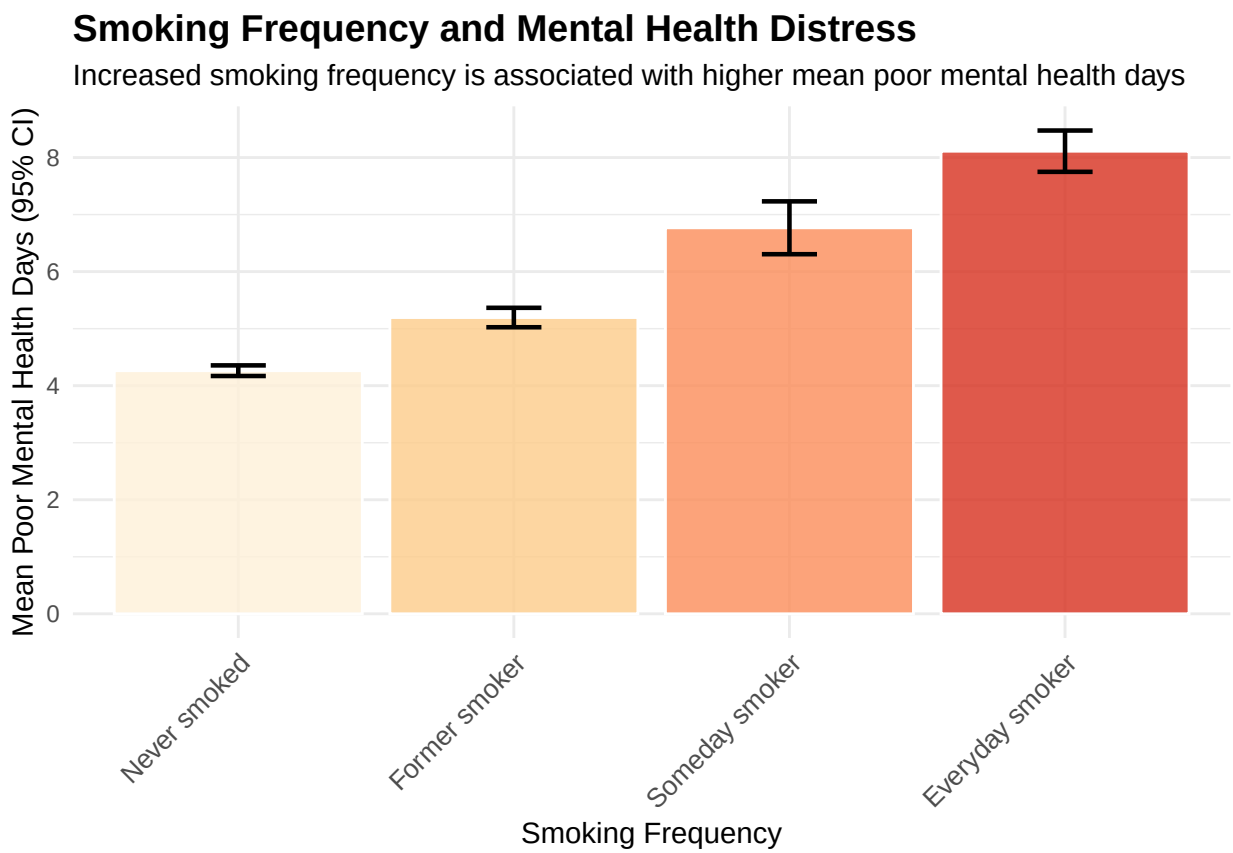
The disparity in mental health between non-smokers and everyday smokers was highly pronounced (t = -20.17), representing the strongest contrast among all smoking categories. Post-hoc pairwise comparisons using Bonferroni adjustment revealed statistically significant differences between all smoking groups ($p < 0.0001$). Everyday smokers had the highest mean number of poor mental health days, followed by someday smokers, former smokers, and never smokers.

```r
library(ggplot2)


plot_data_smoke <- as.data.frame(smoke_emm)
```

```r
c<-ggplot(plot_data_smoke, aes(x = SMOKER3, y = emmean, fill = SMOKER3)) +
  geom_bar(stat = "identity", alpha = 0.8, color = "white") +

  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2, size = 0.8) +

  scale_fill_brewer(palette = "OrRd", direction = 1) +
  labs(
    title = "Smoking Frequency and Mental Health Distress",
    subtitle = "Increased smoking frequency is associated with higher mean poor mental health days",
    x = "Smoking Frequency",
    y = "Mean Poor Mental Health Days (95% CI)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    plot.title = element_text(face = "bold", size = 14)
  )
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```r
print(c)
```




```
ggsave("mean_smoking_plot.png", width = 8, height = 5, dpi = 300)
```

The error bars represent the 95% confidence intervals for the weighted mean of poor mental health days. The lack of overlap between the confidence intervals across all four smoking categories provides visual evidence of the highly significant differences ($p<0.001$) found in our pairwise comparisons.

```
library(emmeans)
library(dplyr)
library(knitr)

# Pairwise comparisons
smoker_pairs <- pairs(smoke_emm, adjust = "bonferroni")

smoker_pairs_df <- as.data.frame(smoker_pairs)

age_pairs_table <- smoker_pairs_df %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, t.ratio, p.value)

kable(
```

```
  age_pairs_table,
  caption = "Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Smokin
  col.names = c("Comparison", "Mean Difference", "SE", "df", "t Ratio", "p-value"),
  align = "lccccc"
)
```

Table 6: Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Smoking Category

Comparison	Mean Difference	SE	df	t Ratio	p-value
Never smoked - Former smoker	-0.932	0.099	221798	-9.403	<0.001
Never smoked - Someday smoker	-2.506	0.241	221798	-10.390	<0.001
Never smoked - Everyday smoker	-3.849	0.191	221798	-20.167	<0.001
Former smoker - Someday smoker	-1.574	0.252	221798	-6.246	<0.001
Former smoker - Everyday smoker	-2.917	0.204	221798	-14.278	<0.001
Someday smoker - Everyday smoker	-1.343	0.300	221798	-4.475	<0.001

```
smoker_pairs_ci <- confint(smoker_pairs)

smoker_pairs_ci_df <- as.data.frame(smoker_pairs_ci)
```

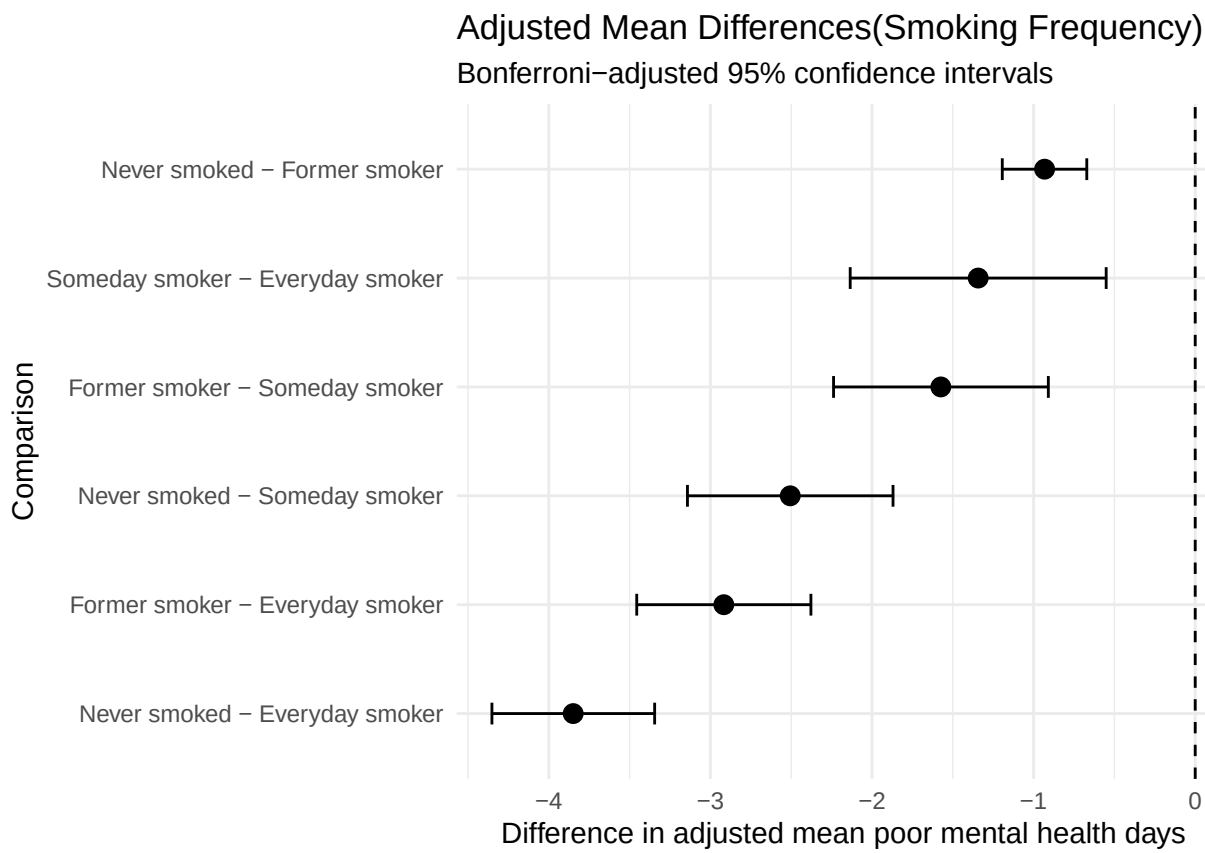
```
sig_smoke_pairs_table_ci <- smoker_pairs_df %>%
  mutate(
    lower.CL = smoker_pairs_ci_df$lower.CL,
    upper.CL = smoker_pairs_ci_df$upper.CL
  ) %>%
  filter(p.value < 0.05) %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    lower.CL = round(lower.CL, 3),
    upper.CL = round(upper.CL, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, lower.CL, upper.CL, t.ratio, p.value)

kable(
  sig_smoke_pairs_table_ci,
  caption = "Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Da
  col.names = c("Comparison", "Mean Difference", "SE", "df", "CI Lower", "CI Upper", "t Ratio", "p-valu
  align = "lcccccc"
)
```

Table 7: Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Smoking Category

Comparison	Mean Difference	SE	df	CI Lower	CI Upper	t Ratio	p-value
Never smoked - Former smoker	-0.932	0.099	221798	-1.194	-0.671	-9.403	<0.001
Never smoked - Someday smoker	-2.506	0.241	221798	-3.142	-1.870	-10.390	<0.001
Never smoked - Everyday smoker	-3.849	0.191	221798	-4.352	-3.345	-20.167	<0.001
Former smoker - Someday smoker	-1.574	0.252	221798	-2.238	-0.909	-6.246	<0.001
Former smoker - Everyday smoker	-2.917	0.204	221798	-3.456	-2.378	-14.278	<0.001
Someday smoker - Everyday smoker	-1.343	0.300	221798	-2.135	-0.551	-4.475	<0.001

Bonferroni adjusted pairwise comparisons show that individuals who smokes everyday report significantly more poor mental health days compared to all other individuals who does not smoke. The differences increase with age, with the largest gap observed between never smoked and everyday smoker.

```r
library(ggplot2)
library(dplyr)

plot_df_smoke <- sig_smoke_pairs_table_ci

a <- ggplot(plot_df_smoke, aes(x = estimate, y = reorder(contrast, estimate))) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = `lower.CL`, xmax = `upper.CL`), height = 0.2) +
  geom_vline(xintercept = 0, linetype = 2) +
  labs(
    x = "Difference in adjusted mean poor mental health days",
    y = "Comparison",
    title = "Adjusted Mean Differences(Smoking Frequency)",
    subtitle = "Bonferroni-adjusted 95% confidence intervals"
  ) +
  theme_minimal()
print(a)
```




```r
ggsave("posthoc_smoking_plot.png", width = 8, height = 5, dpi = 300)
```

All pairwise comparisons were statistically significant (Bonferroni adjusted), with the greatest disparity observed between everyday smokers and those who have never smoked.In our analysis, we didn't just do one test; we compared 6 different pairs of smoking groups. The weighted Wald test indicated a highly significant association between smoking frequency and reported poor mental health days Post-hoc comparisons using the Bonferroni adjustment revealed that all pairwise differences between smoking groups were statistically significant ($p<0.001$). Specifically, 'Everyday smokers' reported the highest mean number of poor mental health days , which was significantly higher than 'Someday smokers','Former smokers' and 'Never smokers'.As shown in the adjusted mean difference plot, the largest disparity was observed between everyday smokers and never-smokers, with the former experiencing nearly four additional days of poor mental health per month."

To complement the analysis of mean differences, we also conducted an ordinal logistic regression to examine the odds of falling into higher categories of poor mental health. This allowed us to assess whether smoking status predicts not just the average number of days, but the severity of mental health distress."

```r
# Ordinal Model for Smoking
smoke_ordinal_model <- svyolr(MENTHLTH_CAT ~ SMOKER3, design = brfss_design)

# Odds Ratios
exp(coef(smoke_ordinal_model))
```

```
##                SMOKER3Former smoker               SMOKER3Someday smoker
##                            1.057463                            1.601409
##              SMOKER3Everyday smoker    0: Resilient|1-13: Low/Moderate
```

```
##                            1.766597                            1.392863
## 1-13: Low/Moderate|14-29: Frequent        14-29: Frequent|30: Chronic
##                            5.944753                           15.647480
```

```r
summary(smoke_ordinal_model)
```

```
## Call:
## svyolr(MENTHLTH_CAT ~ SMOKER3, design = brfss_design)
##
## Coefficients:
##                            Value Std. Error   t value
## SMOKER3Former smoker   0.0558729 0.02244803  2.488988
## SMOKER3Someday smoker  0.4708839 0.04864650  9.679707
## SMOKER3Everyday smoker 0.5690550 0.03784391 15.036897
##
## Intercepts:
##                                    Value    Std. Error t value
## 0: Resilient|1-13: Low/Moderate      0.3314   0.0122    27.0639
## 1-13: Low/Moderate|14-29: Frequent   1.7825   0.0146   122.1123
## 14-29: Frequent|30: Chronic          2.7503   0.0197   139.3200
```

```r
library(dplyr)
library(stringr)

coeffs <- summary(smoke_ordinal_model)$coefficients

smoker_results <- coeffs[str_detect(rownames(coeffs), "^SMOKER3"), , drop = FALSE]

OR <- exp(smoker_results[, "Value"])
CI <- cbind(
  exp(smoker_results[, "Value"] - 1.96 * smoker_results[, "Std. Error"]),
  exp(smoker_results[, "Value"] + 1.96 * smoker_results[, "Std. Error"])
)

raw_p_val <- 2 * pnorm(abs(smoker_results[, "t value"]), lower.tail = FALSE)
adj_p_val <- pmin(raw_p_val * 3, 1)

final_table <- data.frame(
  Smoking_Status = str_remove(rownames(smoker_results), "^SMOKER3"),
  Odds_Ratio = round(OR, 2),
  Lower_95_CI = round(CI[, 1], 2),
  Upper_95_CI = round(CI[, 2], 2),
  P_Value_Adj = format.pval(adj_p_val, eps = .001, digits = 3)
)

print(final_table)
```

```
##                          Smoking_Status Odds_Ratio Lower_95_CI Upper_95_CI
## SMOKER3Former smoker      Former smoker       1.06        1.01        1.11
## SMOKER3Someday smoker    Someday smoker       1.60        1.46        1.76
## SMOKER3Everyday smoker Everyday smoker       1.77        1.64        1.90
##                         P_Value_Adj
## SMOKER3Former smoker         0.0384
```

```
## SMOKER3Someday smoker       <0.001
## SMOKER3Everyday smoker      <0.001
```

```r
rownames(final_table) <- NULL

kable(
  final_table,
  caption = "Odds Ratios for Mental Health Severity by Smoking frequency(Never Smoker reference)",
  col.names = c("Smoking Status", "Odds Ratio", "Lower 95% CI", "Upper 95% CI", "Adjusted p-value"),
  align = "lcccc"
)
```

Table 8: Odds Ratios for Mental Health Severity by Smoking frequency(Never Smoker reference)

Smoking Status	Odds Ratio	Lower 95% CI	Upper 95% CI	Adjusted p-value
Former smoker	1.06	1.01	1.11	0.0384
Someday smoker	1.60	1.46	1.76	<0.001
Everyday smoker	1.77	1.64	1.90	<0.001

An ordinal logistic regression model was conducted to examine the association between smoking status and the severity of poor mental health days . Using 'Never Smokers' as the reference group, a significant dose response relationship was observed. Everyday smokers showed the highest risk, with 1.77 times higher odds (95% CI [1.64, 1.90], Padj <0.001) of falling into a more severe mental health category. Someday smokers also exhibited significantly increased odds (OR = 1.60, 95% CI [1.46, 1.76], Padj<0.001). Notably, even former smokers maintained a small but statistically significant increase in risk (OR = 1.06, 95% CI [1.01, 1.11], 0.0384), even after applying for Bonferroni adjustment for multiple comparisons.

SEX

Null Hypothesis (H0):- The mean number of poor mental health days (MENTHLTH) is the same for females and males.

Alternative Hypothesis(H1):- The mean number of poor mental health days (MENTHLTH) is higher for females than for males.

A significance level of alpha = 0.05 is used.

```r
levels(brfss1$SEX)
```

```
## [1] "Female" "Male"
```

```r
brfss1$SEX <- relevel(brfss1$SEX, ref = "Male")
levels(brfss1$SEX)
```

```
## [1] "Male"   "Female"
```

```r
library(survey)


brfss_design <- svydesign(
  id = ~1,
  weights = ~ LLCPWT,
  data = brfss1
)
```

```r
set.seed(1)
library(survey)
library(dplyr)
library(knitr)


# Weighted means by sex
sex_mean <- svyby(~MENTHLTH, ~SEX, brfss_design, svymean, na.rm = TRUE)

# Confidence intervals
sex_mean_ci <- confint(sex_mean)

sex_results <- data.frame(
  Sex = as.character(sex_mean$SEX),
  Weighted_Mean = round(sex_mean$MENTHLTH, 2),
  Standard_Error = round(sex_mean$se, 3),
  CI_Lower = round(sex_mean_ci[, 1], 2),
  CI_Upper = round(sex_mean_ci[, 2], 2)
)

kable(
  sex_results,
  caption = "Weighted Mean Number of Poor Mental Health Days by Sex",
  col.names = c("Sex", "Weighted Mean", "Standard Error", "CI Lower", "CI Upper"),
  align = "c"
)
```

Table 9: Weighted Mean Number of Poor Mental Health Days by Sex

	Sex	Weighted Mean	Standard Error	CI Lower	CI Upper
Male	Male	4.14	0.055	4.04	4.25
Female	Female	5.51	0.060	5.39	5.62

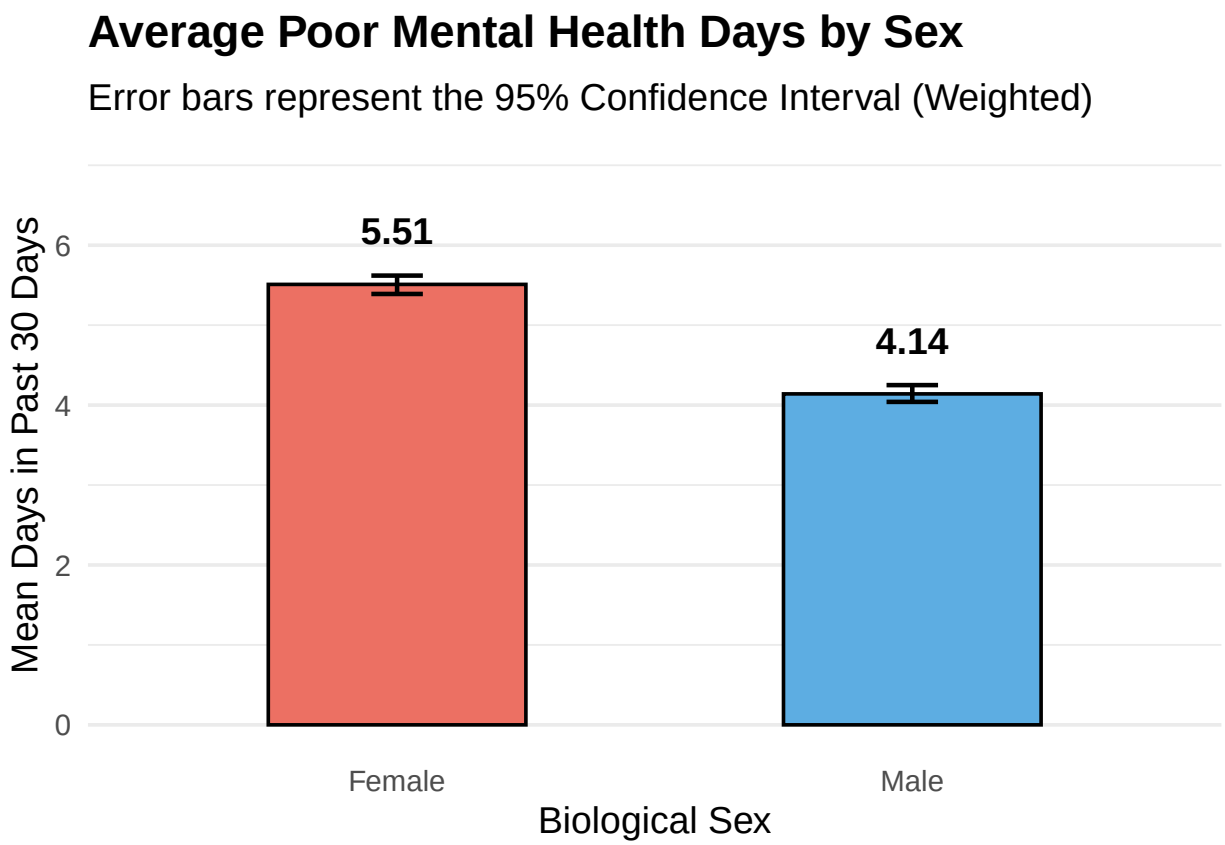
```r
# Survey-weighted t-test
sex_ttest <- svyttest(MENTHLTH ~ SEX, brfss_design,alternative="less")

ttest_results <- data.frame(
  t_Statistic = round(as.numeric(sex_ttest$statistic), 3),
  Degrees_of_Freedom = round(as.numeric(sex_ttest$parameter), 0),
  p_value = format.pval(as.numeric(sex_ttest$p.value), digits = 3, eps = .001),
  Mean_Difference = round(as.numeric(sex_ttest$estimate), 3),
  CI_Lower = round(sex_ttest$conf.int[1], 3),
  CI_Upper = round(sex_ttest$conf.int[2], 3)
)

kable(
  ttest_results,
  caption = "Survey-Weighted t-Test Comparing Mean Poor Mental Health Days by Sex",
  col.names = c("t Statistic", "Degrees of Freedom", "p-value", "Mean Difference", "CI Lower", "CI Uppe
  align = "c"
)
```

Table 10: Survey-Weighted t-Test Comparing Mean Poor Mental Health Days by Sex

t Statistic	Degrees of Freedom	p-value	Mean Difference	CI Lower	CI Upper
16.755	221800	<0.001	1.362	1.203	1.522

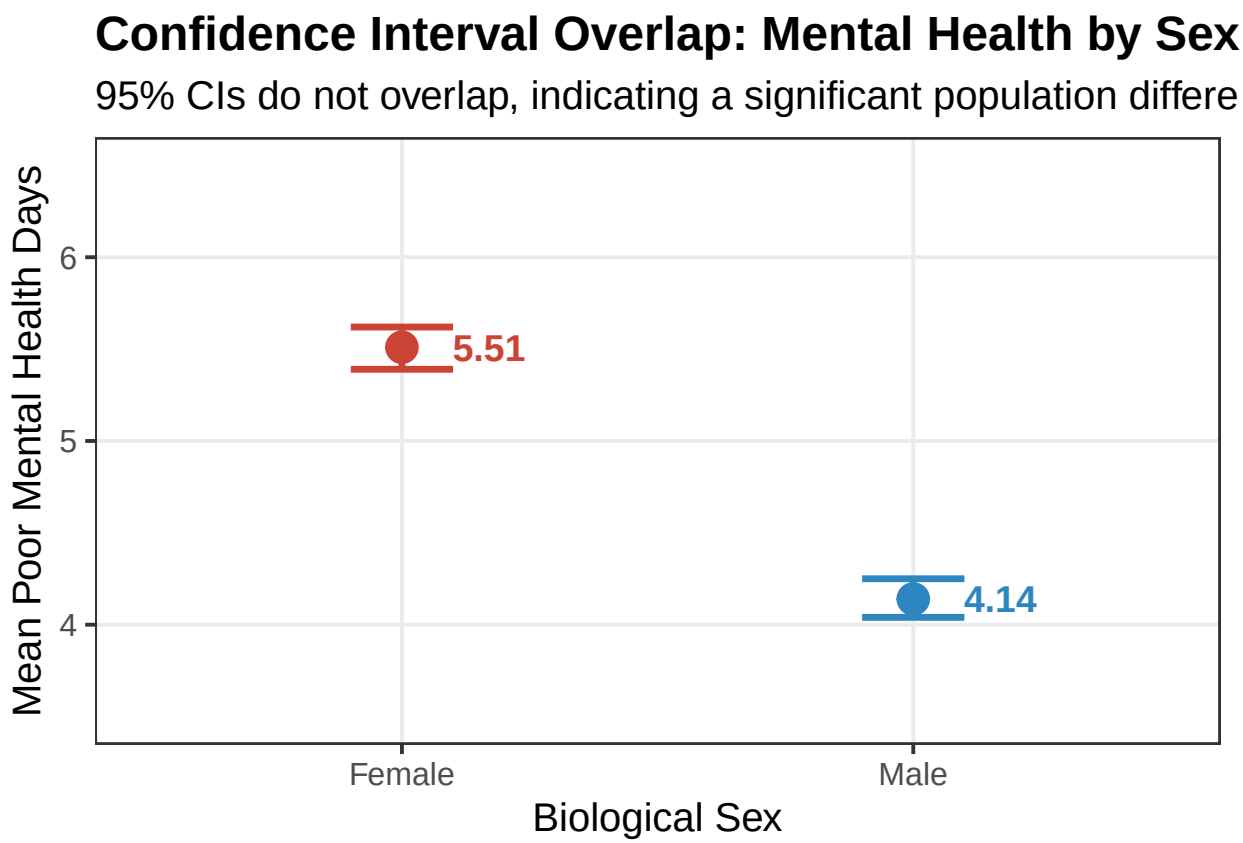
A survey weighted independent samples t-test was conducted to compare the mean number of poor mental health days between biological sexes. The analysis revealed a significant disparity ($t(221800) = 16.76$, $p < 0.001$), with females reporting an average of 1.362 more days of distress per month than males (95 % CI [1.203, 1.522]). These results provide evidence to reject the null hypothesis and support the alternative hypothesis that females experience a significantly higher average burden of poor mental health in the U.S. adult population. The Standard Error shows the precision of our estimate. It tells us how much the mean would likely vary if we drew a different sample of the same size from the population.

```r
ggplot(sex_results, aes(x = Sex, y = Weighted_Mean, fill = Sex)) +
  geom_col(width = 0.5, color = "black", show.legend = FALSE) +
  geom_errorbar(aes(ymin = CI_Lower, ymax = CI_Upper),
                width = 0.1, size = 0.8, color = "black") +

  geom_text(aes(label = Weighted_Mean), vjust = -1.5, fontface = "bold", size = 5) +

  expand_limits(y = 7) +
  theme_minimal(base_size = 14) +
  scale_fill_manual(values = c("Male" = "#5DADE2", "Female" = "#EC7063")) +
  labs(
    title = "Average Poor Mental Health Days by Sex",
    subtitle = "Error bars represent the 95% Confidence Interval (Weighted)",
    x = "Biological Sex",
    y = "Mean Days in Past 30 Days"
  ) +
  theme(
    plot.title = element_text(face = "bold", margin = margin(b = 10)),
    panel.grid.major.x = element_blank()
  )
```




```r
library(ggplot2)

f<- ggplot(sex_results, aes(x = Sex, y = Weighted_Mean, color = Sex)) +

  geom_point(size = 5) +

  geom_errorbar(aes(ymin = CI_Lower, ymax = CI_Upper),
                width = 0.2, size = 1.2) +

  geom_text(aes(label = Weighted_Mean),
            hjust = -0.7, fontface = "bold", size = 5) +

  theme_bw(base_size = 15) +
  scale_color_manual(values = c("Male" = "#2E86C1", "Female" = "#CB4335")) +
  expand_limits(y = c(3.5, 6.5)) +
  labs(
    title = "Confidence Interval Overlap: Mental Health by Sex",
    subtitle = "95% CIs do not overlap, indicating a significant population difference",
    x = "Biological Sex",
    y = "Mean Poor Mental Health Days"
  ) +
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold")
  )
```

```r
print(f)
```




```r
ggsave("mean_sex_plot.png", width = 8, height = 5, dpi = 300)
```

The visual analysis of 95% Confidence Intervals further reinforces the t test results. The lack of any overlap between the female (5.51 days) and male (4.14 days) intervals provides clear evidence of a significant population level disparity. Furthermore, the narrow width of the error bars indicates high precision in our estimates, confirming that females experience a consistently higher burden of poor mental health across the U.S. adult population.

To further see the results, we performed **chi square test** to see the association between mental health Category and Sex

Null Hypothesis (H0): Mental health categories are independent of biological sex. (The distribution across Resilient, Low, Frequent, and Chronic is proportionally the same for men and women). Alternative Hypothesis (H1): Mental health categories are associated with biological sex. (Women are disproportionately represented in higher-distress categories).

```r
#survey-weighted Chi-Square test
sex_cat_chisq <- svychisq(~MENTHLTH_CAT + SEX, design = brfss_design)
print(sex_cat_chisq)
```

```
##
##  Pearson's X^2: Rao & Scott adjustment
##
```

```
## data:  svychisq(~MENTHLTH_CAT + SEX, design = brfss_design)
## F = 167.87, ndf = 2.9952e+00, ddf = 6.6433e+05, p-value < 2.2e-16
```

A survey-weighted Chi-Square test showed a significant association between sex and mental health category. So we rejected H0

```r
sex_cat_props <- svyby(~MENTHLTH_CAT, ~SEX, brfss_design, svymean, na.rm = TRUE)
print(sex_cat_props)
```

```
##           SEX MENTHLTH_CAT0: Resilient MENTHLTH_CAT1-13: Low/Moderate
## Male     Male                0.6194949                      0.2520499
## Female Female                0.5148193                      0.3082754
##        MENTHLTH_CAT14-29: Frequent MENTHLTH_CAT30: Chronic
## Male                     0.0727347              0.05572052
## Female                   0.1036764              0.07322893
##        se.MENTHLTH_CAT0: Resilient se.MENTHLTH_CAT1-13: Low/Moderate
## Male                   0.003331202                       0.002997957
## Female                 0.003476367                       0.003204649
##        se.MENTHLTH_CAT14-29: Frequent se.MENTHLTH_CAT30: Chronic
## Male                      0.001737131                0.001512706
## Female                    0.002090741                0.001735142
```

```r
sex_table_final <- as.data.frame(sex_cat_props)

colnames(sex_table_final) <- c(
  "Sex",
  "Resilient (0 days)", "Low/Mod (1-13 days)",
  "Frequent (14-29 days)", "Chronic (30 days)",
  "SE_Resilient", "SE_LowMod", "SE_Frequent", "SE_Chronic"
)

sex_table_final[, 2:5] <- round(sex_table_final[, 2:5] * 100, 1)

sex_table_final[, 6:9] <- round(sex_table_final[, 6:9], 4)

library(knitr)
kable(
  sex_table_final[, 1:5],
  caption = "Distribution of Mental Health Severity by Sex (Weighted Percentages)",
  align = "c",
  row.names = FALSE
)
```

Table 11: Distribution of Mental Health Severity by Sex (Weighted Percentages)

Sex	Resilient (0 days)	Low/Mod (1-13 days)	Frequent (14-29 days)	Chronic (30 days)
Male	61.9	25.2	7.3	5.6
Female	51.5	30.8	10.4	7.3

```r
categories <- c("1-13: Low/Moderate", "14-29: Frequent", "30: Chronic")
```

```r
for (cat in categories) {

  sub_design <- subset(brfss_design, MENTHLTH_CAT %in% c("0: Resilient", cat))


  sub_design <- update(sub_design, is_target = (MENTHLTH_CAT == cat))

  model <- svyglm(is_target ~ SEX, design = sub_design, family = quasibinomial)

  print(paste("Likelihood for", cat, "vs Resilient:"))
  print(exp(coef(model))["SEXFemale"])
}
```

```
## [1] "Likelihood for 1-13: Low/Moderate vs Resilient:"
## SEXFemale
##  1.471754
## [1] "Likelihood for 14-29: Frequent vs Resilient:"
## SEXFemale
##  1.715225
## [1] "Likelihood for 30: Chronic vs Resilient:"
## SEXFemale
##  1.581432
```

```r
gender_risk_table <- data.frame(
  Distress_Category = c("Low/Moderate (1-13 days)", "Frequent (14-29 days)", "Chronic (30 days)"),
  Odds_Ratio = c(1.47, 1.72, 1.58)
)

gender_risk_table$Risk_Increase <- paste0((gender_risk_table$Odds_Ratio - 1) * 100, "% higher odds")

print(gender_risk_table)
```

```
##          Distress_Category Odds_Ratio   Risk_Increase
## 1 Low/Moderate (1-13 days)       1.47 47% higher odds
## 2    Frequent (14-29 days)       1.72 72% higher odds
## 3        Chronic (30 days)       1.58 58% higher odds
```

1.47 (Low/Moderate): Compared to being "Resilient," females have 1.47 times the odds (or are 47% more likely) of being in the Low/Moderate group than males. 1.71 (Frequent): Females have 1.71 times the odds (or are 71% more likely) of experiencing Frequent distress than being Resilient, compared to males. 1.58 (Chronic): Females have 1.58 times the odds (or are 58% more likely) of experiencing Chronic distress than being Resilient, compared to males.

Physical Activity

Null Hypothesis (H0):- The mean number of poor mental health days for those who engage in physical activity is greater than or equal to those who do not.

Alternative Hypothesis(H1):- The mean number of poor mental health days for those who engage in physical activity is significantly lower than those who do not.

A significance level of alpha = 0.05 is used, and a directional survey weighted t test is conducted.

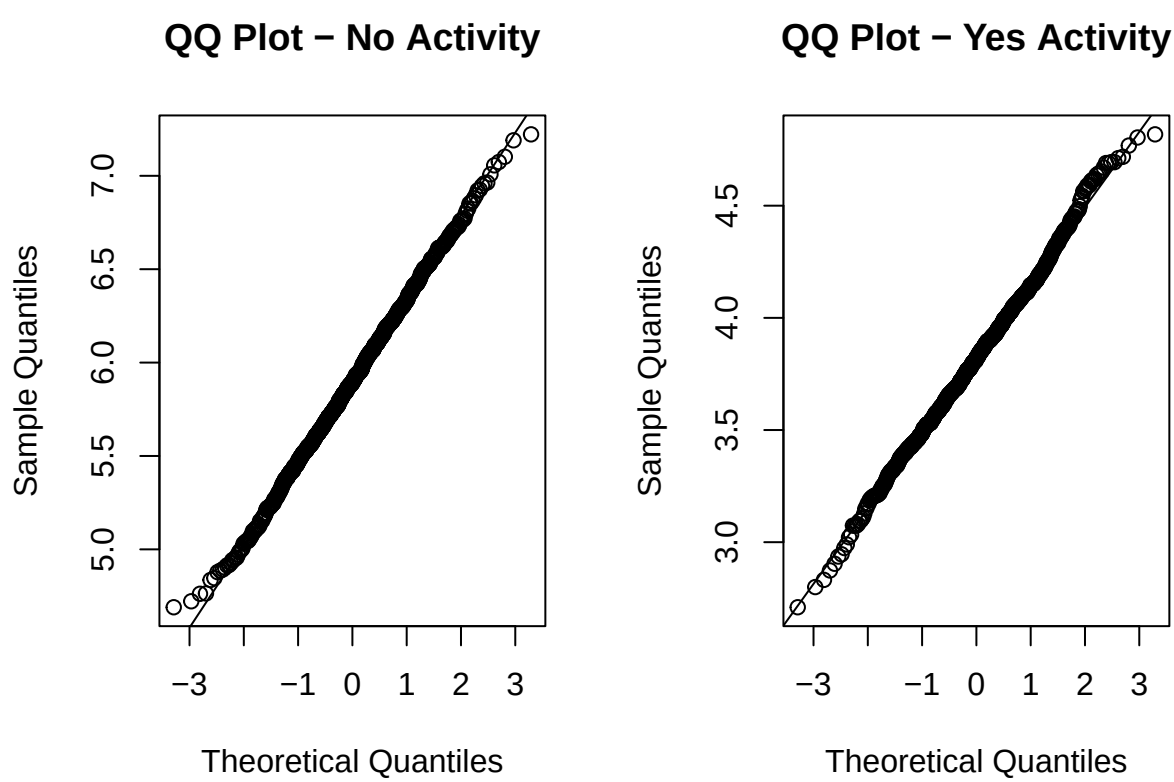
Although the distribution of poor mental health days was right-skewed, a survey-weighted t-test was used to compare group means. Given the large sample size, the Central Limit Theorem ensures that the sampling distribution of the mean is approximately normal. This assumption was further supported by resampling analysis, which showed that the distribution of sample means was approximately normal. Therefore, the use of the weighted t-test is appropriate.

```
no_data  <- brfss1 %>% filter(EXERANY2 == "No") %>% pull(MENTHLTH)
yes_data <- brfss1 %>% filter(EXERANY2 == "Yes") %>% pull(MENTHLTH)
```

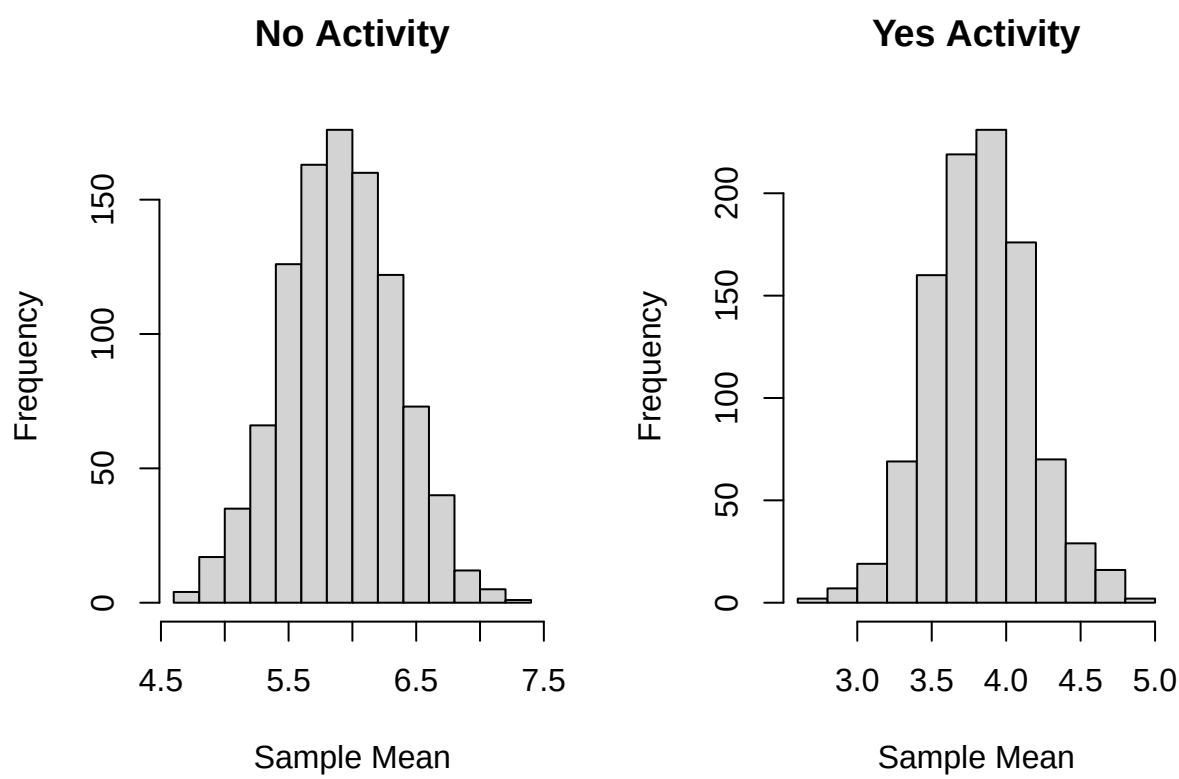
```
sample_means <- function(data, n = 1000, size = 500) {
  replicate(n, mean(sample(data, size, replace = TRUE), na.rm = TRUE))
}
```

```
no_means  <- sample_means(no_data)
yes_means <- sample_means(yes_data)
```

```
par(mfrow = c(1,2))

qqnorm(no_means, main = "QQ Plot - No Activity")
qqline(no_means)

qqnorm(yes_means, main = "QQ Plot - Yes Activity")
qqline(yes_means)
```



```r
par(mfrow = c(1,2))

hist(no_means, main = "No Activity", xlab = "Sample Mean")

hist(yes_means, main = "Yes Activity", xlab = "Sample Mean")
```



To visually confirm the normality of the sampling distribution, we generated histograms of the simulated sample means (Figure X). Both the 'No Activity' and 'Yes Activity' groups produced symmetrical, bell-shaped distributions. This visual evidence, combined with the earlier QQ plots, provides a high degree of confidence that the Central Limit Theorem is satisfied, ensuring the validity of our survey-weighted t-test."

```r
set.seed(1)

library(survey)

library(dplyr)

library(knitr)


brfss1$EXERANY2 <- factor(brfss1$EXERANY2, levels = c("No", "Yes"))

brfss1$MENTHLTH <- as.numeric(brfss1$MENTHLTH)
```

```r
activity_mean <- svyby(~MENTHLTH, ~EXERANY2, brfss_design, svymean, na.rm = TRUE)

# Confidence intervals

activity_mean_ci <- confint(activity_mean)


activity_results <- data.frame(

  Physical_Activity = as.character(activity_mean$EXERANY2),

  Weighted_Mean = round(activity_mean$MENTHLTH, 2),

  Standard_Error = round(activity_mean$se, 3),

  CI_Lower = round(activity_mean_ci[, 1], 2),

  CI_Upper = round(activity_mean_ci[, 2], 2)

)


kable(

  activity_results,

  caption = "Weighted Mean Number of Poor Mental Health Days by Physical Activity",

  col.names = c("Physical Activity", "Weighted Mean", "Standard Error", "CI Lower", "CI Upper"),

  align = "c"

)
```

Table 12: Weighted Mean Number of Poor Mental Health Days by Physical Activity

	Physical Activity	Weighted Mean	Standard Error	CI Lower	CI Upper
No	No	6.22	0.105	6.02	6.43
Yes	Yes	4.45	0.043	4.37	4.54

```r
# Directed t-test

activity_ttest <- svyttest(MENTHLTH ~ EXERANY2, brfss_design,alternative = "greater")



ttest_results <- data.frame(

  t_Statistic = round(as.numeric(activity_ttest$statistic), 3),

  Degrees_of_Freedom = round(as.numeric(activity_ttest$parameter), 0),
```

```
  p_value = format.pval(as.numeric(activity_ttest$p.value), digits = 3, eps = .001),

  Mean_Difference = round(as.numeric(activity_ttest$estimate), 3),

  CI_Lower = round(activity_ttest$conf.int[1], 3),

  CI_Upper = round(activity_ttest$conf.int[2], 3)

)



kable(

  ttest_results,

  caption = "Directed Survey-Weighted t-Test:if Physically Inactive Adults report more Poor Mental Heal

  col.names = c("t Statistic", "Degrees of Freedom", "p-value", "Mean Difference", "CI Lower", "CI Uppe

  align = "c",
  row.names = FALSE

)
```

Table 13: Directed Survey-Weighted t-Test:if Physically Inactive Adults report more Poor Mental Health Days

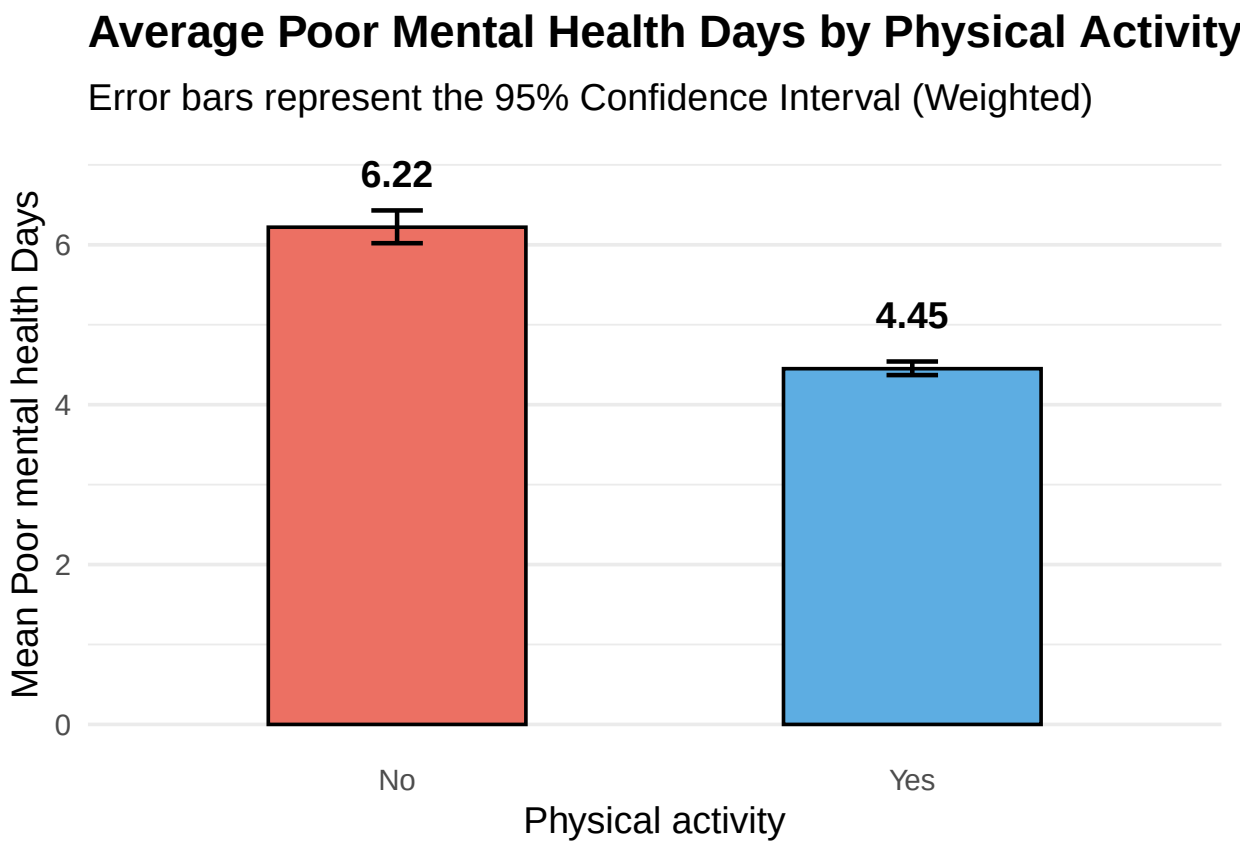
t Statistic	Degrees of Freedom	p-value	Mean Difference	CI Lower	CI Upper
-15.663	221800	<0.001	-1.772	-1.994	-1.55

Since the p-value is less than 0.05, we reject the null hypothesis and conclude that mean poor mental health days is more among the individual who are not physically active. Physical activity is associated with fewer poor mental health days in the U.S. adult population.

Physically active individuals had significantly fewer poor mental health days than physically inactive individuals, with a mean difference of 1.77 days. Equivalently, the weighted mean difference for active versus inactive individuals was -1.77 days (95% CI: -1.99 to -1.55, $p < 0.001$).

```
g<- ggplot(activity_results, aes(x = Physical_Activity, y = Weighted_Mean, fill = Physical_Activity)) +
  geom_col(width = 0.5, color = "black", show.legend = FALSE) +
  geom_errorbar(aes(ymin = CI_Lower, ymax = CI_Upper),
                width = 0.1, size = 0.8, color = "black") +

  geom_text(aes(label = Weighted_Mean), vjust = -1.5, fontface = "bold", size = 5) +

  expand_limits(y = 7) +
  theme_minimal(base_size = 14) +
  scale_fill_manual(values = c("Yes" = "#5DADE2", "No" = "#EC7063")) +
  labs(
    title = "Average Poor Mental Health Days by Physical Activity",
    subtitle = "Error bars represent the 95% Confidence Interval (Weighted)",
```

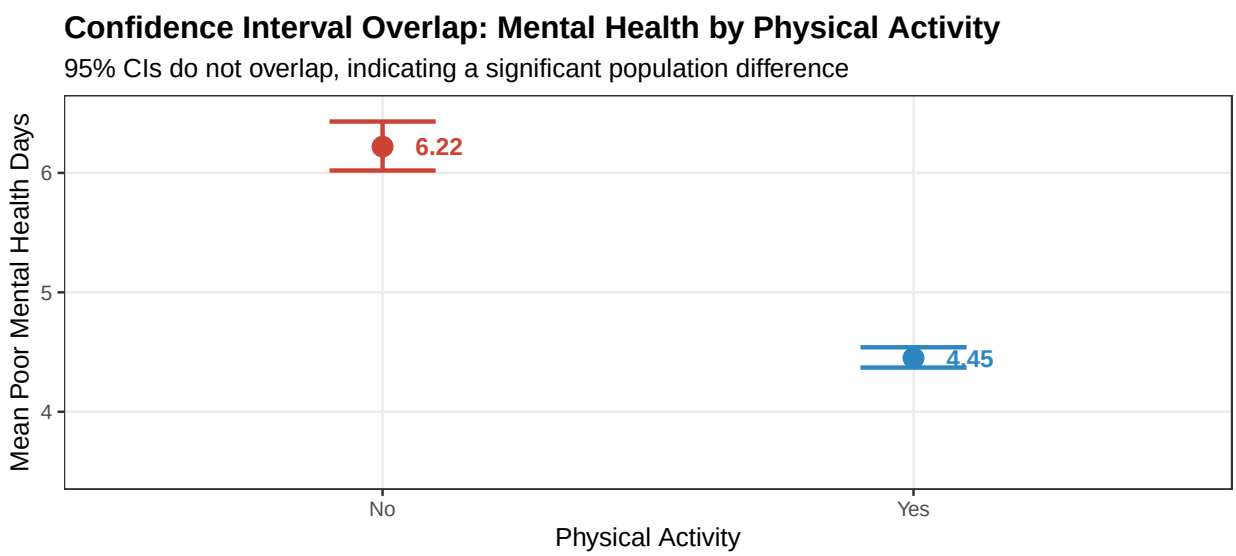
```
    x = "Physical activity",
    y = "Mean Poor mental health Days"
  ) +
  theme(
    plot.title = element_text(face = "bold", margin = margin(b = 10)),
    panel.grid.major.x = element_blank()
  )
print(g)
```




```
ggsave("mean_physicalactivity_plot.png", width = 8, height = 5, dpi = 300)
```

```
library(ggplot2)

ggplot(activity_results, aes(x = Physical_Activity, y = Weighted_Mean, color = Physical_Activity)) +

  geom_point(size = 5) +

  geom_errorbar(aes(ymin = CI_Lower, ymax = CI_Upper),
                width = 0.2, size = 1.2) +

  geom_text(aes(label = Weighted_Mean),
            hjust = -0.7, fontface = "bold", size = 5) +

  theme_bw(base_size = 15) +
  scale_color_manual(values = c("Yes" = "#2E86C1", "No" = "#CB4335")) +
```

```r
  expand_limits(y = c(3.5, 6.5)) +
  labs(
    title = "Confidence Interval Overlap: Mental Health by Physical Activity",
    subtitle = "95% CIs do not overlap, indicating a significant population difference",
    x = "Physical Activity",
    y = "Mean Poor Mental Health Days"
  )+
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold")
  )
```



Null Hypothesis : There is no association between physical activity and the severity of mental health distress, after adjusting for age.

Alternative Hypothesis : Engaging in physical activity is significantly associated with lower odds of being in a more severe mental health category,independent of age.

```r
brfss_design$variables$EXERANY2 <- factor(brfss_design$variables$EXERANY2, levels = c("No", "Yes"))

exer_ordinal_model <- svyolr(MENTHLTH_CAT ~ EXERANY2 + AGE4, design = brfss_design)

summary_exer <- summary(exer_ordinal_model)
exer_OR <- exp(coef(exer_ordinal_model))
exer_CI <- exp(confint(exer_ordinal_model))

print(exer_OR)
```

```
##                        EXERANY2Yes                        AGE435-49
##                          0.7782062                        0.7052538
##                          AGE450-64                          AGE465+
##                          0.4878164                        0.2537105
##    0: Resilient|1-13: Low/Moderate 1-13: Low/Moderate|14-29: Frequent
##                          0.6210162                        2.8157308
```

```
##        14-29: Frequent|30: Chronic
##                          7.4851739
```

```
exer_coefs <- as.data.frame(coef(summary(exer_ordinal_model)))

exer_coefs$OR <- exp(exer_coefs$Value)

exer_final_table <- exer_coefs %>%
  filter(row_number() <= 4) %>%
  mutate(
    `Odds Ratio` = round(OR, 2),
    `t value` = round(`t value`, 2),
  ) %>%
  select(`Odds Ratio`, `t value`)

print(exer_final_table)
```

```
##             Odds Ratio t value
## EXERANY2Yes       0.78   -9.49
## AGE435-49         0.71  -13.40
## AGE450-64         0.49  -26.63
## AGE465+           0.25  -50.70
```

```
regTermTest(exer_ordinal_model, "EXERANY2")
```

```
## Wald test for EXERANY2
##  in svyolr(MENTHLTH_CAT ~ EXERANY2 + AGE4, design = brfss_design)
## F =  89.99146  on  1  and  221797  df: p= < 2.22e-16
```

After adjusting for age, the ordinal logistic regression revealed that engaging in physical activity has a significant protective effect. Individuals who engaged in physical activity in the past 30 days exhibited 22% lower odds of falling into a more severe mental health category compared to inactive individuals. So we reject the null hypothesis.

Loneliness

Null Hypthesis (H0) : The mean number of poor mental health days is equal across all loneliness groups. Alternative Hypothesis(HA): At least one loneliness group has a different mean number of poor mental health days.

```
model_sdlonely <- svyglm(MENTHLTH ~ SDLONELY, design = brfss_design)
regTermTest(model_sdlonely, "SDLONELY")
```

```
## Wald test for SDLONELY
##  in svyglm(formula = MENTHLTH ~ SDLONELY, design = brfss_design)
## F =  1419.007  on  4  and  221797  df: p= < 2.22e-16
```

The Wald test indicates that F(4, 221797) = 1419.01, $p < 0.001$. Reject H0, mean number of poor mental health daysdiffers across loneliness levels.

Post-hoc pairwise comparisons using Bonferroni adjustment:-

```r
library(emmeans)

lonely_emm <- emmeans(model_sdlonely, ~ SDLONELY)
lonely_emm
```

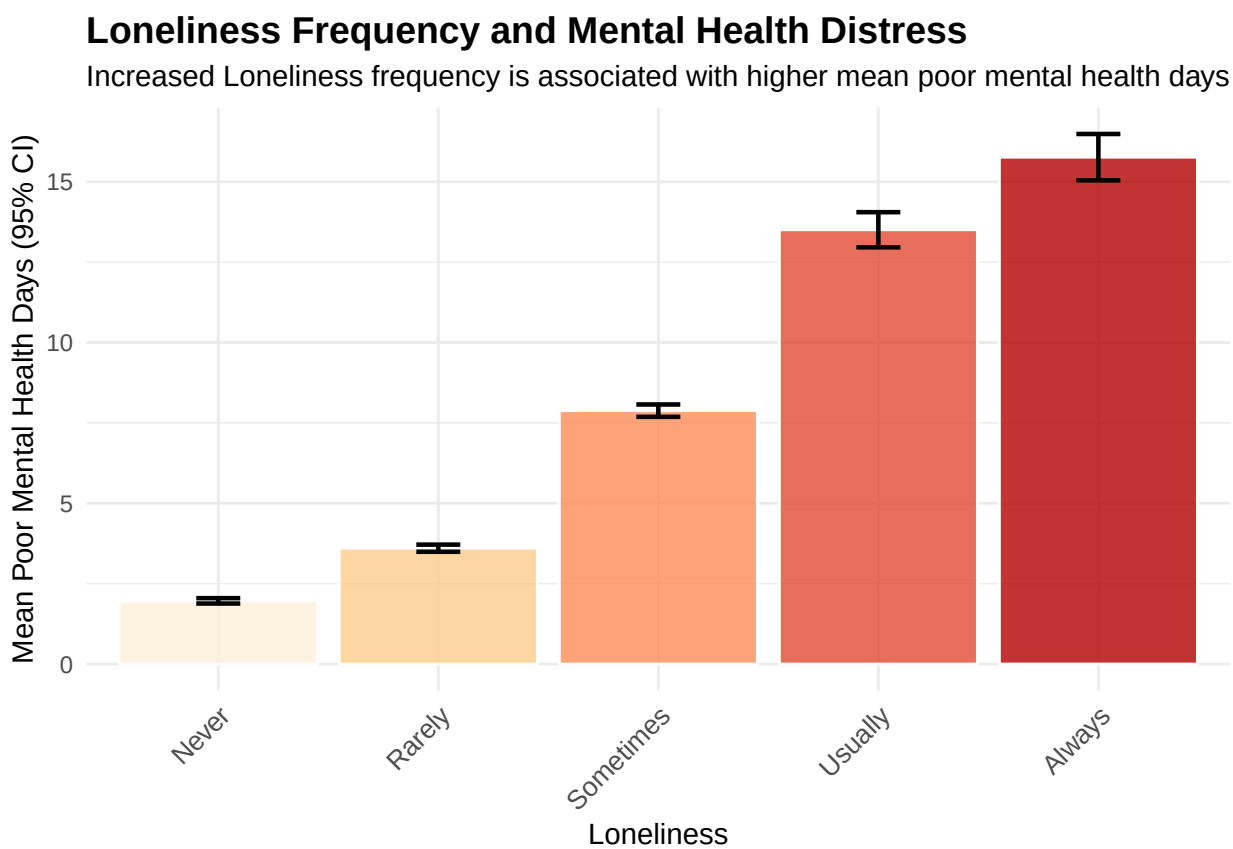
```
##  SDLONELY  emmean     SE     df lower.CL upper.CL
##  Never       1.97 0.0427 221797     1.88     2.05
##  Rarely      3.60 0.0565 221797     3.49     3.71
##  Sometimes   7.88 0.0980 221797     7.69     8.07
##  Usually    13.51 0.2790 221797    12.96    14.05
##  Always     15.77 0.3680 221797    15.05    16.49
##
## Confidence level used: 0.95
```

```r
pairs(lonely_emm, adjust = "bonferroni")
```

```
##  contrast            estimate     SE     df t.ratio p.value
##  Never - Rarely         -1.64 0.0708 221797 -23.120 <0.0001
##  Never - Sometimes      -5.92 0.1070 221797 -55.329 <0.0001
##  Never - Usually       -11.54 0.2820 221797 -40.894 <0.0001
##  Never - Always        -13.80 0.3700 221797 -37.295 <0.0001
##  Rarely - Sometimes     -4.28 0.1130 221797 -37.800 <0.0001
##  Rarely - Usually       -9.90 0.2850 221797 -34.793 <0.0001
##  Rarely - Always       -12.16 0.3720 221797 -32.705 <0.0001
##  Sometimes - Usually    -5.63 0.2960 221797 -19.025 <0.0001
##  Sometimes - Always     -7.88 0.3800 221797 -20.726 <0.0001
##  Usually - Always       -2.26 0.4610 221797  -4.894 <0.0001
##
## P value adjustment: bonferroni method for 10 tests
```

```r
library(ggplot2)

plot_data_lonely <- as.data.frame(lonely_emm)


d<- ggplot(plot_data_lonely, aes(x = SDLONELY, y = emmean, fill = SDLONELY)) +
  geom_bar(stat = "identity", alpha = 0.8, color = "white") +

  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2, size = 0.8) +

  scale_fill_brewer(palette = "OrRd", direction = 1) +
  labs(
    title = "Loneliness Frequency and Mental Health Distress",
    subtitle = "Increased Loneliness frequency is associated with higher mean poor mental health days",
    x = "Loneliness",
    y = "Mean Poor Mental Health Days (95% CI)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    plot.title = element_text(face = "bold", size = 14)
```

```
  )
print(d)
```




```
ggsave("mean_lonely_plot.png", width = 8, height = 5, dpi = 300)
```

```
library(emmeans)
library(dplyr)
library(knitr)

# Pairwise comparisons
lonely_pairs <- pairs(lonely_emm, adjust = "bonferroni")

lonely_pairs_df <- as.data.frame(lonely_pairs)

lonely_pairs_table <- lonely_pairs_df %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, t.ratio, p.value)

kable(
  lonely_pairs_table,
```

```
  caption = "Posthoc Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Loneliness",
  col.names = c("Loneliness Levels Comparison", "Mean Difference", "SE", "df", "t Ratio", "p-value"),
  align = "lccccc"
)
```

Table 14: Posthoc Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Loneliness

Loneliness Levels Comparison	Mean Difference	SE	df	t Ratio	p-value
Never - Rarely	-1.638	0.071	221797	-23.121	<0.001
Never - Sometimes	-5.916	0.107	221797	-55.329	<0.001
Never - Usually	-11.542	0.282	221797	-40.894	<0.001
Never - Always	-13.800	0.370	221797	-37.295	<0.001
Rarely - Sometimes	-4.278	0.113	221797	-37.800	<0.001
Rarely - Usually	-9.905	0.285	221797	-34.793	<0.001
Rarely - Always	-12.163	0.372	221797	-32.705	<0.001
Sometimes - Usually	-5.626	0.296	221797	-19.025	<0.001
Sometimes - Always	-7.884	0.380	221797	-20.726	<0.001
Usually - Always	-2.258	0.461	221797	-4.894	<0.001

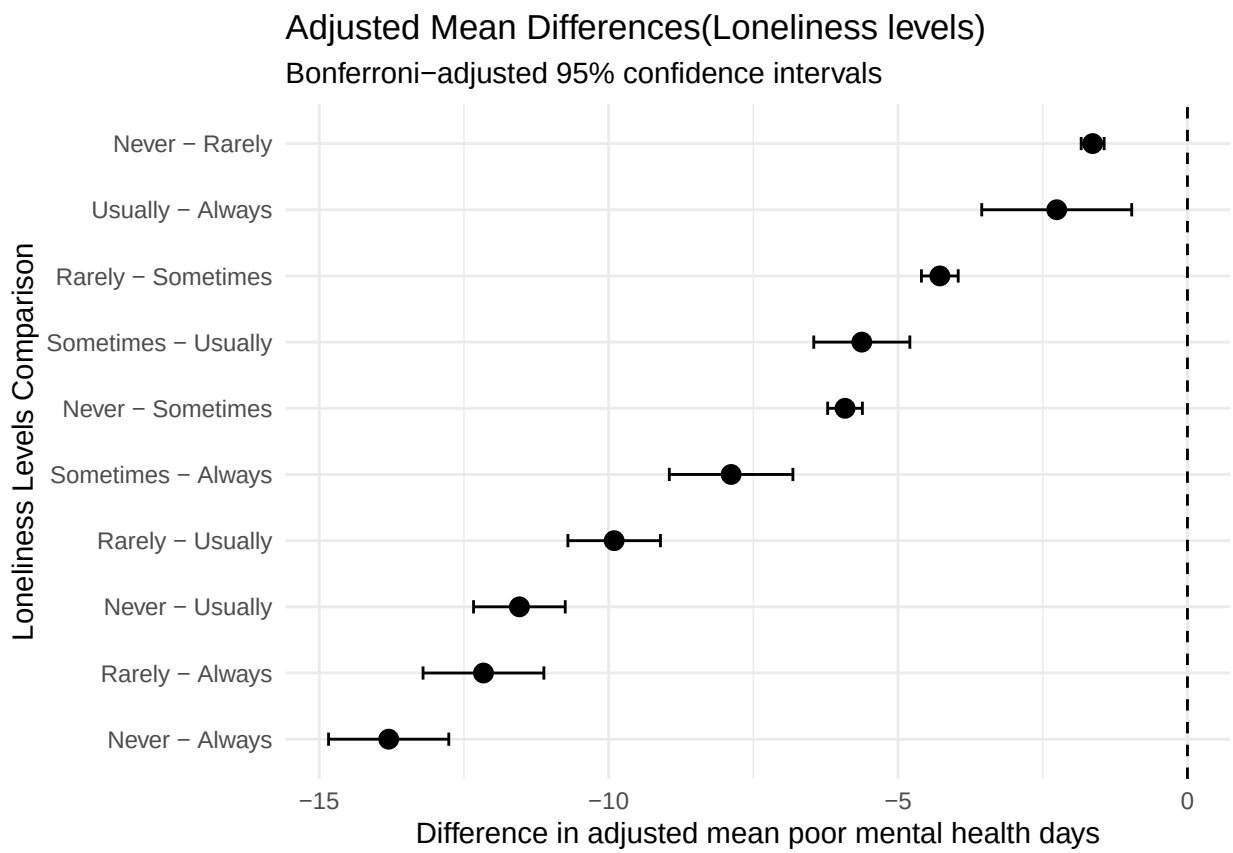
```
lonely_pairs_ci <- confint(lonely_pairs)

lonely_pairs_ci_df <- as.data.frame(lonely_pairs_ci)
```

```
sig_lonely_pairs_table_ci <- lonely_pairs_df %>%
  mutate(
    lower.CL = lonely_pairs_ci_df$lower.CL,
    upper.CL = lonely_pairs_ci_df$upper.CL
  ) %>%
  filter(p.value < 0.05) %>%
  mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    lower.CL = round(lower.CL, 3),
    upper.CL = round(upper.CL, 3),
    t.ratio = round(t.ratio, 3),
    p.value = format.pval(p.value, digits = 3, eps = .001)
  ) %>%
  select(contrast, estimate, SE, df, lower.CL, upper.CL, t.ratio, p.value)

kable(
  sig_lonely_pairs_table_ci,
  caption = "Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Da
  col.names = c("Comparison", "Mean Difference", "SE", "df", "CI Lower", "CI Upper", "t Ratio", "p-valu
  align = "lcccccc"
)
```

Table 15: Significant Bonferroni-Adjusted Pairwise Comparisons of Adjusted Mean Poor Mental Health Days by Loneliness

Comparison	Mean Difference	SE	df	CI Lower	CI Upper	t Ratio	p-value
Never - Rarely	-1.638	0.071	221797	-1.836	-1.439	-23.121	<0.001
Never - Sometimes	-5.916	0.107	221797	-6.216	-5.616	-55.329	<0.001
Never - Usually	-11.542	0.282	221797	-12.334	-10.750	-40.894	<0.001
Never - Always	-13.800	0.370	221797	-14.839	-12.762	-37.295	<0.001
Rarely - Sometimes	-4.278	0.113	221797	-4.596	-3.961	-37.800	<0.001
Rarely - Usually	-9.905	0.285	221797	-10.704	-9.105	-34.793	<0.001
Rarely - Always	-12.163	0.372	221797	-13.207	-11.119	-32.705	<0.001
Sometimes - Usually	-5.626	0.296	221797	-6.456	-4.796	-19.025	<0.001
Sometimes - Always	-7.884	0.380	221797	-8.952	-6.817	-20.726	<0.001
Usually - Always	-2.258	0.461	221797	-3.554	-0.963	-4.894	<0.001

```r
library(ggplot2)
library(dplyr)

plot_df_smoke <- sig_lonely_pairs_table_ci

e<- ggplot(plot_df_smoke, aes(x = estimate, y = reorder(contrast, estimate))) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = `lower.CL`, xmax = `upper.CL`), height = 0.2) +
  geom_vline(xintercept = 0, linetype = 2) +
  labs(
    x = "Difference in adjusted mean poor mental health days",
    y = "Loneliness Levels Comparison",
    title = "Adjusted Mean Differences(Loneliness levels)",
    subtitle = "Bonferroni-adjusted 95% confidence intervals"
  ) +
  theme_minimal()
print(e)
```




```
ggsave("post_hoc_lonely_plot.png", width = 8, height = 5, dpi = 300)
```

The figure displays the Bonferroni-adjusted 95% confidence intervals for the differences in mean poor mental health days between loneliness levels. All comparisons were statistically significant, as evidenced by all confidence intervals remaining entirely to the left of the zero null line. The most pronounced difference was observed between the 'Never' and 'Always' lonely groups (Mean Difference -13.8 days). These results demonstrate a significant relationship, where every increase in the frequency of loneliness corresponds to a statistically significant increase in the number of reported poor mental health days.

We also conducted an ordinal logistic regression to examine the odds of falling into higher categories of poor mental health. This allowed us to assess whether loneliness predicts not just the average number of days, but the severity of mental health distress.

Null Hypothesis (H0): There is no relationship between loneliness frequency and mental health category. The odds of being in a more severe mental health category (e.g., Chronic) versus a less severe one (e.g., Resilient) are equal across all loneliness frequency groups.

Alternative Hypothesis (H1): Loneliness frequency is a significant predictor of mental health category. The odds of being in a more severe mental health category significantly differ based on how often an individual feels lonely.

```
sdlonely_ordinal_model <- svyolr(MENTHLTH_CAT ~ SDLONELY, design = brfss_design)

exp(coef(sdlonely_ordinal_model))
```

```
##                    SDLONELYRarely                    SDLONELYSometimes
```

```
##                          2.580477                          6.194657
##                   SDLONELYUsually                     SDLONELYAlways
##                         15.710114                         21.649780
##    0: Resilient|1-13: Low/Moderate 1-13: Low/Moderate|14-29: Frequent
##                          3.444092                         18.376926
##        14-29: Frequent|30: Chronic
##                         53.308500
```

```r
summary(sdlonely_ordinal_model)
```

```
## Call:
## svyolr(MENTHLTH_CAT ~ SDLONELY, design = brfss_design)
##
## Coefficients:
##                       Value Std. Error  t value
## SDLONELYRarely    0.9479741 0.02528915 37.48540
## SDLONELYSometimes 1.8236871 0.02726354 66.89106
## SDLONELYUsually   2.7543047 0.05350410 51.47839
## SDLONELYAlways    3.0749953 0.07325887 41.97437
##
## Intercepts:
##                                    Value    Std. Error t value
## 0: Resilient|1-13: Low/Moderate      1.2367   0.0193    64.0033
## 1-13: Low/Moderate|14-29: Frequent   2.9111   0.0238   122.2481
## 14-29: Frequent|30: Chronic          3.9761   0.0281   141.6039
```

```r
coeffs <- summary(sdlonely_ordinal_model)$coefficients
sdlonely_results <- coeffs[1:4, ]
```

```r
sdlonely_results
```

```
##                       Value Std. Error  t value
## SDLONELYRarely    0.9479741 0.02528915 37.48540
## SDLONELYSometimes 1.8236871 0.02726354 66.89106
## SDLONELYUsually   2.7543047 0.05350410 51.47839
## SDLONELYAlways    3.0749953 0.07325887 41.97437
```

```r
coefs <- sdlonely_results[1:4, "Value"]
ses <- sdlonely_results[1:4, "Std. Error"]
t_vals <- sdlonely_results[1:4, "t value"]

OR <- exp(coefs)
lower_CI <- exp(coefs - 1.96 * ses)
upper_CI <- exp(coefs + 1.96 * ses)

raw_p_val <- 2 * pnorm(abs(t_vals), lower.tail = FALSE)
adj_p_val <- raw_p_val * 4  # Adjusted for the 4 loneliness levels
adj_p_val <- ifelse(adj_p_val > 1, 1, adj_p_val) # Cap at 1.0


final_table <- data.frame(
  "Loneliness_Level" = c("Rarely", "Sometimes", "Usually", "Always"),
```

```r
  "Odds_Ratio" = round(OR, 2),
  "Lower_95_CI" = round(lower_CI, 2),
  "Upper_95_CI" = round(upper_CI, 2),
  "P_value_Adj" = format.pval(adj_p_val, eps = .001, digits = 3)
)

print(final_table)
```

```
##                    Loneliness_Level Odds_Ratio Lower_95_CI Upper_95_CI
## SDLONELYRarely               Rarely       2.58        2.46        2.71
## SDLONELYSometimes         Sometimes       6.19        5.87        6.53
## SDLONELYUsually             Usually      15.71       14.15       17.45
## SDLONELYAlways               Always      21.65       18.75       24.99
##                    P_value_Adj
## SDLONELYRarely          <0.001
## SDLONELYSometimes       <0.001
## SDLONELYUsually         <0.001
## SDLONELYAlways          <0.001
```

```r
library(knitr)

rownames(final_table) <- NULL

kable(
  final_table,
  caption = "Odds Ratios for Mental Health Severity by Loneliness Level",
  col.names = c("Loneliness Level", "Odds Ratio", "Lower 95% CI", "Upper 95% CI", "Adjusted p-value"),
  align = "lcccc"
)
```

Table 16: Odds Ratios for Mental Health Severity by Loneliness Level

Loneliness Level	Odds Ratio	Lower 95% CI	Upper 95% CI	Adjusted p-value
Rarely	2.58	2.46	2.71	<0.001
Sometimes	6.19	5.87	6.53	<0.001
Usually	15.71	14.15	17.45	<0.001
Always	21.65	18.75	24.99	<0.001

This ordinal logistic regression shows that loneliness frequency is a powerful predictor of mental health category severity, with all coefficients highly significant ($p < 0.001$). Using "Never" lonely as the reference category, the coefficients increase progressively: "Rarely" (2.58), "Sometimes" (6.19), "Usually" (15.71), and "Always" (21.65). This means that individuals who report feeling lonely "Always" have 21.65 higher odds of being in a more severe mental health category (e.g., Chronic vs. Frequent) compared to those who "Never" feel lonely. The intercepts(in the summary) represent thresholds between categories, with the largest gap between the Chronic threshold and the Frequent threshold (3.98), indicating a substantial distinction between frequent and chronic mental health distress. Overall, the model demonstrates a clear, relationship where more frequent loneliness corresponds to increasingly worse mental health outcomes.

Do Social Determinants explain variation in poor mental health days beyond demographic/behavioral factors :-

Null Hypothesis :- Social determinants (loneliness, emotional support, life satisfaction, and perceived safety) are not associated with poor mental health after controlling for demographic and behavioral factors.

Mathematically: The coefficients for SDLONELY ,EMTSUPRT ,Support,Life Satisfaction are equal to zero in the full model.

Alternative Hypothesis: Social determinants (Loneliness, Emotional Support,Life Satisfaction, Environment Safety) are associated with poor mental health after controlling for demographic and behavioral factors.

Mathematically: At least one social determinant coefficient is significantly different from zero in the full model

A joint Wald test was conducted to assess whether the set of social determinant variables (loneliness, emotional support, life satisfaction, and perceived safety) contributes significantly to the model after adjusting for demographic and behavioral covariates. The null hypothesis states that all coefficients corresponding to the social determinants are equal to zero, while the alternative hypothesis states that at least one of these coefficients is non-zero.

```
model_full <- svyolr(
  MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 + EXERANY2 + SDLONELY +EMTSUPRT + LSATISFY + HOWS
  design = brfss_design
)
```

```
# Joint Wald Test
# This tests the "Social Block" while holding everything else constant
social_beyond_test <- regTermTest(model_full, ~ SDLONELY + EMTSUPRT+LSATISFY+HOWSAFE1)

print(social_beyond_test)
```

```
## Wald test for SDLONELY EMTSUPRT LSATISFY HOWSAFE1
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  563.2536  on  14  and  221766  df: p= < 2.22e-16
```

```
regTermTest(model_full, ~ SDLONELY)
```

```
## Wald test for SDLONELY
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  534.088  on  4  and  221766  df: p= < 2.22e-16
```

```
regTermTest(model_full, ~ EMTSUPRT)
```

```
## Wald test for EMTSUPRT
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  78.46932  on  4  and  221766  df: p= < 2.22e-16
```

```
regTermTest(model_full , ~LSATISFY)
```

```
## Wald test for LSATISFY
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  426.5093  on  3  and  221766  df: p= < 2.22e-16
```

```r
regTermTest(model_full , ~HOWSAFE1)
```

```
## Wald test for HOWSAFE1
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  11.59131  on  3  and  221766  df: p= 1.3616e-07
```

To test whether social determinants provide explanatory power beyond demographic and behavioral factors, a joint Wald test was performed on the fully adjusted model. The results were highly significant ($p<0.001$), supporting the hypothesis that social factors are independent and vital predictors of mental health. Individual Wald tests revealed that Loneliness (SDLONELY) was the most influential factor ($F = 534.088$), followed by Life Satisfaction. This suggests that while demographics like income and age matter, a person's subjective social experience is the primary driver of mental health distress in this population.

```r
vars_to_test <- c("AGE4", "SEX", "INCOMG1", "RACE", "SMOKER3",
                  "EXERANY2", "SDLONELY", "EMTSUPRT", "LSATISFY", "HOWSAFE1")

results_list <- list()

for (v in vars_to_test) {
  test_formula <- as.formula(paste0("~", v))

  test_output <- tryCatch(
    regTermTest(model_full, test_formula),
    error = function(e) NULL
  )

  if (!is.null(test_output)) {
    results_list[[v]] <- data.frame(
      Variable = v,
      F_Statistic = as.numeric(test_output$Ftest),
      p_value = as.numeric(test_output$p)
    )
  }
}

library(dplyr)
library(knitr)

final_leaderboard <- bind_rows(results_list) %>%
  mutate(
    F_Statistic = round(F_Statistic, 2),
    p_value = format.pval(p_value, eps = .001, digits = 3)
  ) %>%
  arrange(desc(F_Statistic))

kable(final_leaderboard, caption = "Significance of each factor(Wald Test)", align = "c")
```

Table 17: Significance of each factor(Wald Test)

Variable	F_Statistic	p_value
SEX	653.79	<0.001
SDLONELY	534.09	<0.001
AGE4	526.55	<0.001
LSATISFY	426.51	<0.001
EMTSUPRT	78.47	<0.001
SMOKER3	67.70	<0.001
RACE	36.07	<0.001
HOWSAFE1	11.59	<0.001
INCOMG1	5.58	<0.001
EXERANY2	2.64	0.104

```r
model_results <- as.data.frame(summary(model_full)$coefficients)

#Odds Ratios and 95% Confidence Intervals
model_results$OR <- exp(model_results$Value)
model_results$Lower_CI <- exp(model_results$Value - (1.96 * model_results$`Std. Error`))
model_results$Upper_CI <- exp(model_results$Value + (1.96 * model_results$`Std. Error`))

or_summary <- model_results %>%
  select(OR, Lower_CI, Upper_CI, `t value`) %>%
  mutate(across(everything(), round, 2))
```

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘across(everything(), round, 2)‘.
## Caused by warning:
## ! The ‘...‘ argument of ‘across()‘ is deprecated as of dplyr 1.1.0.
## Supply arguments directly to ‘.fns‘ through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```r
print(or_summary)
```

```
##                                                  OR Lower_CI Upper_CI t value
## AGE435-49                                      0.78     0.74     0.82   -8.89
## AGE450-64                                      0.55     0.52     0.58  -19.92
## AGE465+                                        0.29     0.27     0.31  -38.51
## SEXFemale                                      1.72     1.65     1.79   25.57
## INCOMG1$15-25k                                 0.93     0.83     1.04   -1.31
## INCOMG1$25-35k                                 0.95     0.85     1.05   -0.98
## INCOMG1$35-50k                                 0.97     0.88     1.08   -0.49
## INCOMG1$50k-$100k                              1.04     0.94     1.15    0.80
## INCOMG1$100k-$200k                             1.13     1.02     1.25    2.32
## INCOMG1>$200k                                  1.11     0.98     1.24    1.67
## RACEAsian                                      0.61     0.50     0.75   -4.79
```

```
## RACEBlack                                        0.85     0.72     1.00   -2.00
## RACEHispanic                                     0.62     0.53     0.73   -5.65
## RACEMultiracial                                  1.17     0.96     1.42    1.58
## RACENative Hawaiian or other Pacific\nIslander  0.96     0.66     1.41   -0.19
## RACEOther                                        0.91     0.70     1.18   -0.69
## RACEWhite                                        1.01     0.86     1.18    0.08
## SMOKER3Former smoker                             1.32     1.26     1.39   11.06
## SMOKER3Someday smoker                            1.37     1.23     1.52    5.84
## SMOKER3Everyday smoker                           1.51     1.40     1.63   10.53
## EXERANY2Yes                                      0.95     0.90     1.01   -1.63
## SDLONELYRarely                                   1.90     1.80     2.00   23.08
## SDLONELYSometimes                                3.47     3.27     3.69   40.11
## SDLONELYUsually                                  6.04     5.41     6.75   32.03
## SDLONELYAlways                                   7.44     6.45     8.59   27.41
## EMTSUPRTUsually                                  1.38     1.32     1.46   12.56
## EMTSUPRTSometimes                                1.68     1.57     1.79   15.12
## EMTSUPRTRarely                                   1.89     1.71     2.10   12.22
## EMTSUPRTNever                                    1.34     1.20     1.49    5.15
## LSATISFYSatisfied                                1.78     1.70     1.87   24.22
## LSATISFYDissatisfied                             5.65     5.08     6.28   32.02
## LSATISFYVery dissatisfied                        9.18     7.24    11.63   18.33
## HOWSAFE1Safe                                     1.01     0.97     1.05    0.47
## HOWSAFE1Unsafe                                   1.31     1.18     1.45    5.13
## HOWSAFE1Extremely Unsafe                         1.39     1.12     1.71    3.02
## 0: Resilient|1-13: Low/Moderate                  3.79     3.14     4.59   13.73
## 1-13: Low/Moderate|14-29: Frequent              24.52    20.21    29.76   32.41
## 14-29: Frequent|30: Chronic                     78.83    64.77    95.94   43.58
```

```r
library(dplyr)
library(tibble)
library(knitr)

important_or <- model_results %>%
  rownames_to_column("Variable") %>%
  filter(Variable %in% c(
    "SEXFemale",
    "AGE465+",
    "SMOKER3Everyday smoker",
    "SDLONELYAlways",
    "EMTSUPRTRarely",
    "LSATISFYVery dissatisfied",
    "HOWSAFE1Extremely Unsafe",
    "EXERANY2Yes"
  )) %>%
  mutate(
    Predictor = c(
      "Female vs Male",
      "Age 65+ vs 18-34",
      "Everyday smoker vs Never smoked",
      "Always lonely vs Never",
      "Rare emotional support vs Always",
      "Very dissatisfied vs Very satisfied",
      "Extremely unsafe vs Very safe",
      "Physical activity: Yes vs No"
```

```r
    ),
    `Odds Ratio` = round(OR, 2),
    `Lower 95% CI` = round(Lower_CI, 2),
    `Upper 95% CI` = round(Upper_CI, 2)
  ) %>%
  select(Predictor, `Odds Ratio`, `Lower 95% CI`, `Upper 95% CI`)

kable(
  important_or,
  caption = "Key Adjusted Odds Ratios from the Full Ordinal Regression Model",
  align = "lccc"
)
```

Table 18: Key Adjusted Odds Ratios from the Full Ordinal Regression Model

Predictor	Odds Ratio	Lower 95% CI	Upper 95% CI
Female vs Male	0.29	0.27	0.31
Age 65+ vs 18–34	1.72	1.65	1.79
Everyday smoker vs Never smoked	1.51	1.40	1.63
Always lonely vs Never	0.95	0.90	1.01
Rare emotional support vs Always	7.44	6.45	8.59
Very dissatisfied vs Very satisfied	1.89	1.71	2.10
Extremely unsafe vs Very safe	9.18	7.24	11.63
Physical activity: Yes vs No	1.39	1.12	1.71

The joint Wald tests show that, in the full multivariable ordinal regression, almost all variables are significantly associated with worse mental health. Ranked by overall contribution, the strongest factors are sex, loneliness, age group, and life satisfaction. Emotional support and smoking status are also important, while perceived safety and income have smaller but still significant overall effects. Physical activity is the only variable here that is not statistically significant overal (p=0.104). From the odds ratios, the clearest patterns are: Compared with ages 18-34, older groups have lower odds of being in a worse mental health category. The odds are especially low for adults aged 65+ (OR=0.29), showing a strong age gradient. Females have substantially higher odds of worse mental health than males (OR=1.72). Smoking is associated with poorer mental health, and the odds increase with smoking frequency: former smokers (OR=1.32), someday smokers (OR=1.37), and everyday smokers(OR=1.51). Loneliness shows one of the strongest patterns in the model. Compared with those who never feel lonely, the odds rise sharply for rarely (OR=1.90), sometimes (OR=3.47), usually (OR=6.04), and always lonely (OR=7.44). Life satisfaction is another major predictor. Compared with those who are very satisfied, the odds are much higher for satisfied (OR=1.78), dissatisfied (OR=5.65), and especially very dissatisfied(OR=9.18). Lower emotional support is also associated with worse mental health, with odds ratios ranging from 1.34 to 1.89 relative to always receiving support. Perceived safety has a smaller but meaningful effect: feeling unsafe(OR=1.31) or extremely unsafe (OR=1.39) is associated with worse mental health. Income has a comparatively weak and mixed adjusted effect in this full model. Race shows some significant differences for certain groups, but the effects are generally smaller than those for loneliness, life satisfaction, sex, and age.

In the fully adjusted ordinal regression model, social determinants remained strongly associated with poor mental health after controlling for demographic and behavioral factors. Among all predictors, loneliness and life satisfaction showed the strongest associations, followed by sex and age group. Lower emotional support and smoking were also important contributors, whereas physical activity was not statistically significant in the full model. Overall, these findings suggest that social, behavioral factors play a major role in explaining poor mental health beyond demographic characteristics.

```r
regTermTest(model_full, ~ SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1)
```

```
## Wald test for SDLONELY EMTSUPRT LSATISFY HOWSAFE1
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  563.2536  on  14  and  221766  df: p= < 2.22e-16
```

```r
regTermTest(model_full, ~ AGE4+SEX+INCOMG1+RACE)
```

```
## Wald test for AGE4 SEX INCOMG1 RACE
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  135.8079  on  17  and  221766  df: p= < 2.22e-16
```

```r
regTermTest(model_full, ~ EXERANY2 + SMOKER3)
```

```
## Wald test for EXERANY2 SMOKER3
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + INCOMG1 + RACE + SMOKER3 +
##     EXERANY2 + SDLONELY + EMTSUPRT + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  51.72446  on  4  and  221766  df: p= < 2.22e-16
```

To determine whether social health provides explanatory power beyond demographic and behavioral factors, we conducted joint Wald tests on three distinct blocks of predictors within the fully adjusted model. The results show that while all blocks were highly significant (p<0.001), the social determinants block exhibited the highest statistical strength (563.2536). This was substantially higher than the F-statistics for the demographic (F=135.8079) and behavioral (51.72446) blocks, confirming that social health is the primary determinant of mental health burden in this model.

Does emotional support modify the relationship between loneliness and mental health severity?

Null hypothesis (H 0): The association between loneliness and mental health does not differ by emotional support. Alternative hypothesis (H 1): The association between loneliness and mental health differs by emotional support.

```r
interaction_model_support <- svyolr(
  MENTHLTH_CAT ~ AGE4 + SEX + SDLONELY*EMTSUPRT+INCOMG1 + RACE + SMOKER3 + EXERANY2  + LSATISFY + HOWSA
  design = brfss_design
)
```

```r
regTermTest(interaction_model_support, ~ SDLONELY:EMTSUPRT)
```

```
## Wald test for SDLONELY:EMTSUPRT
##  in svyolr(MENTHLTH_CAT ~ AGE4 + SEX + SDLONELY * EMTSUPRT + INCOMG1 +
##     RACE + SMOKER3 + EXERANY2 + LSATISFY + HOWSAFE1, design = brfss_design)
## F =  4.439064  on  16  and  221750  df: p= 6.6201e-09
```

To assess whether emotional support modifies the relationship between loneliness and mental health, an interaction term between loneliness (SDLONELY) and emotional support (EMTSUPRT) was included in the survey-weighted ordered logistic regression model. The Wald test for the interaction was statistically significant ($F = 4.43$, $p < 0.001$) . This provides strong evidence that the effect of loneliness on chronic poor mental health varies depending on the level of emotional support.

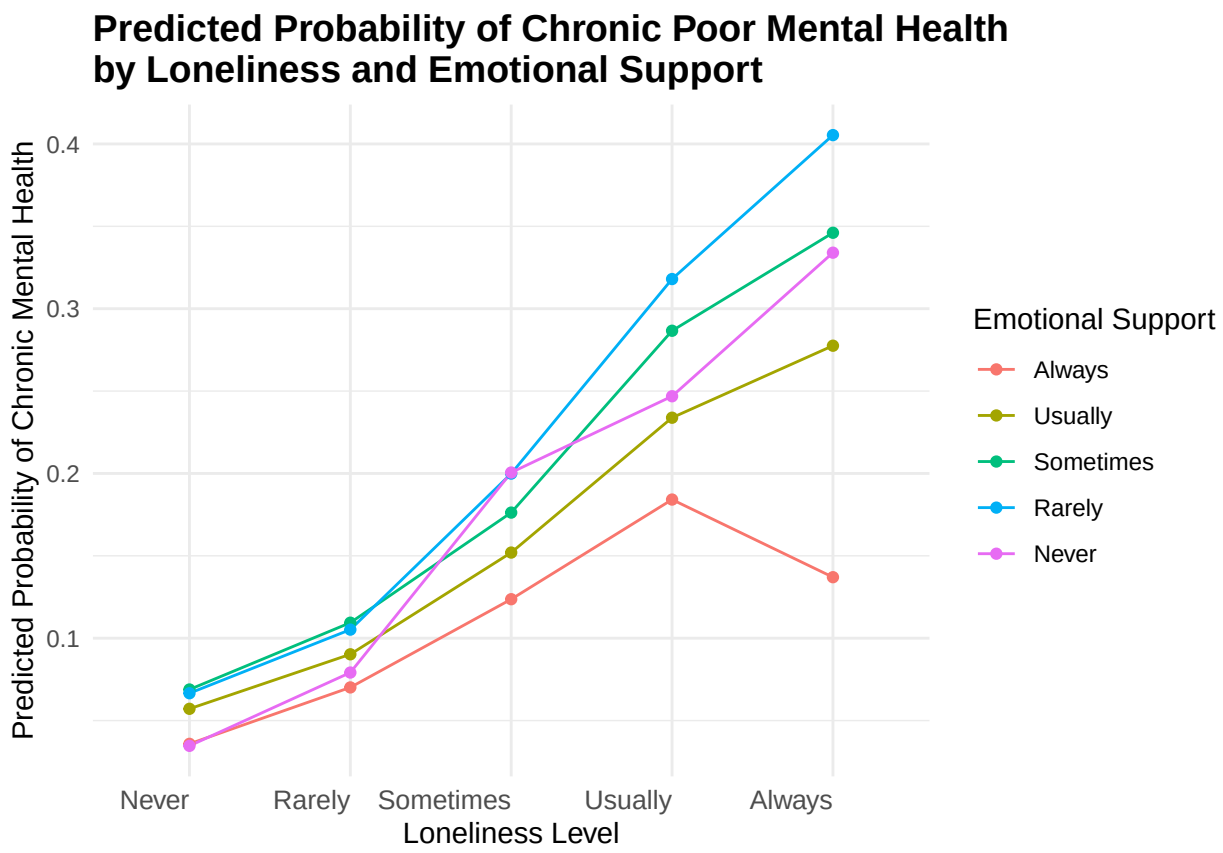
Emotional support significantly moderates the relationship between loneliness and mental health. This provides evidence for the Buffering Effect, where the presence of high social support can help neutralize the negative impact of loneliness on mental health severity.

```
library(emmeans)

emm_prob_support <- emmeans(
  interaction_model_support,
  ~ MENTHLTH_CAT | SDLONELY + EMTSUPRT,
  mode = "prob",
  nuisance = c("SEX", "INCOMG1", "AGE4", "LSATISFY", "HOWSAFE1","RACE","SMOKER3","EXERANY2")
)

emm_prob_support_df <- as.data.frame(emm_prob_support)
```

```
emm_prob_support_chronic <- subset(emm_prob_support_df, MENTHLTH_CAT == 4)
```

Adjusted predicted probabilities were estimated from the interaction model to visualize the relationship between loneliness, emotional support, and chronic poor mental health. After averaging over the remaining variables, the predicted probability of being in the chronic mental health category was plotted across loneliness levels for each emotional support group. The plot shows that chronic poor mental health becomes more likely as loneliness increases, and that this increase is strongest among individuals with low emotional support, indicating a buffering role of emotional support.Predicted probabilities were averaged over the distribution of other covariates in the model, meaning that the results represent population-level estimates rather than values for a specific subgroup.

```
library(ggplot2)

emm_prob_support_chronic$SDLONELY <- factor(
  emm_prob_support_chronic$SDLONELY,
  levels = c("Never", "Rarely", "Sometimes", "Usually", "Always")
)

pred<- ggplot(
  emm_prob_support_chronic,
  aes(x = SDLONELY, y = prob, color = EMTSUPRT, group = EMTSUPRT)
) +
  geom_line() +
  geom_point() +
  labs(
    title = "Predicted Probability of Chronic Poor Mental Health\nby Loneliness and Emotional Support",
    x = "Loneliness Level",
    y = "Predicted Probability of Chronic Mental Health",
    color = "Emotional Support"
  ) +
  theme_minimal()+
    theme(
    axis.text.x = element_text( hjust = 1, size = 10),
    plot.title = element_text(face = "bold", size = 14),

  )
print(pred)
```




```
ggsave("predicted_probability.png", width = 8, height = 5, dpi = 300)
```

The predicted probability plot demonstrates a strong interaction between loneliness and emotional support. While the probability of chronic poor mental health increases with loneliness across all groups, this increase is substantially steeper among individuals with lower emotional support. In contrast, individuals with high emotional support show a much weaker increase, suggesting that emotional support may buffer the adverse effects of loneliness on mental health.

The interaction plot confirms the buffering effect of emotional support on loneliness related mental health distress. While the predicted probability of chronic poor mental health increases with loneliness for all groups, the magnitude of this increase is significantly mitigated for individuals who report having emotional support 'Always' or 'Usually' available. Specifically, for those who are 'Always' lonely, having constant emotional support reduces the predicted probability of chronic distress by more than half compared to those with limited support.

Our hypothesis was strongly supported. Not only did loneliness and social support independently predict mental health, but their interaction was highly significant ($p<0.001$). The analysis reveals a clear buffering effect, where high emotional support significantly reduces the probability of chronic mental health distress, even among the most lonely individuals

Use case in real life These findings suggest that interventions aimed at increasing emotional support may help reduce the negative mental health impact of loneliness, particularly among individuals who experience high levels of loneliness.

```
library(dplyr)
library(survey)
library(knitr)
```

```r
vars <- c("AGE4", "INCOMG1", "SMOKER3", "SDLONELY","RACE")

summary_list <- lapply(vars, function(var) {

  unweighted <- brfss1 %>%
    count(.data[[var]]) %>%
    mutate(
      Proportion = n / sum(n),
      Type = "Unweighted"
    )

  weighted <- svytable(as.formula(paste0("~", var)), brfss_design) %>%
    prop.table() %>%
    as.data.frame()

  colnames(weighted) <- c(var, "Proportion")
  weighted$Type <- "Weighted"

  bind_rows(
    unweighted %>% rename(Level = !!var),
    weighted %>% rename(Level = !!var)
  ) %>%
    mutate(Variable = var) %>%
    select(Variable, Level, Type, Proportion)
})

final_summary <- bind_rows(summary_list)

kable(final_summary,
      caption = "Comparison of Weighted vs Unweighted Distributions Across Key Variables",
      digits = 3)
```

Table 19: Comparison of Weighted vs Unweighted Distributions Across Key Variables

Variable	Level	Type	Proportion
AGE4	18-34	Unweighted	0.171
AGE4	35-49	Unweighted	0.194
AGE4	50-64	Unweighted	0.244
AGE4	65+	Unweighted	0.390
AGE4	18-34	Weighted	0.294
AGE4	35-49	Weighted	0.236
AGE4	50-64	Weighted	0.234
AGE4	65+	Weighted	0.235
INCOMG1	<$15k	Unweighted	0.058
INCOMG1	$15-25k	Unweighted	0.091
INCOMG1	$25-35k	Unweighted	0.114
INCOMG1	$35-50k	Unweighted	0.137
INCOMG1	$50k-$100k	Unweighted	0.305
INCOMG1	$100k-$200k	Unweighted	0.218
INCOMG1	>$200k	Unweighted	0.076
INCOMG1	<$15k	Weighted	0.068

Variable	Level	Type	Proportion
INCOMG1	$15-25k	Weighted	0.099
INCOMG1	$25-35k	Weighted	0.116
INCOMG1	$35-50k	Weighted	0.132
INCOMG1	$50k-$100k	Weighted	0.279
INCOMG1	$100k-$200k	Weighted	0.216
INCOMG1	>$200k	Weighted	0.090
SMOKER3	Never smoked	Unweighted	0.616
SMOKER3	Former smoker	Unweighted	0.276
SMOKER3	Someday smoker	Unweighted	0.033
SMOKER3	Everyday smoker	Unweighted	0.075
SMOKER3	Never smoked	Weighted	0.656
SMOKER3	Former smoker	Weighted	0.239
SMOKER3	Someday smoker	Weighted	0.035
SMOKER3	Everyday smoker	Weighted	0.070
SDLONELY	Never	Unweighted	0.378
SDLONELY	Rarely	Unweighted	0.325
SDLONELY	Sometimes	Unweighted	0.236
SDLONELY	Usually	Unweighted	0.037
SDLONELY	Always	Unweighted	0.025
SDLONELY	Never	Weighted	0.366
SDLONELY	Rarely	Weighted	0.314
SDLONELY	Sometimes	Weighted	0.247
SDLONELY	Usually	Weighted	0.044
SDLONELY	Always	Weighted	0.029
RACE	American Indian/Alaskan Native	Unweighted	0.018
RACE	Asian	Unweighted	0.027
RACE	Black	Unweighted	0.089
RACE	Hispanic	Unweighted	0.108
RACE	Multiracial	Unweighted	0.024
RACE	Native Hawaiian or other Pacific		
Islander	Unweighted	0.004	
RACE	Other	Unweighted	0.007
RACE	White	Unweighted	0.724
RACE	American Indian/Alaskan Native	Weighted	0.012
RACE	Asian	Weighted	0.073
RACE	Black	Weighted	0.120
RACE	Hispanic	Weighted	0.203
RACE	Multiracial	Weighted	0.026
RACE	Native Hawaiian or other Pacific		
Islander	Weighted	0.004	
RACE	Other	Weighted	0.009
RACE	White	Weighted	0.553